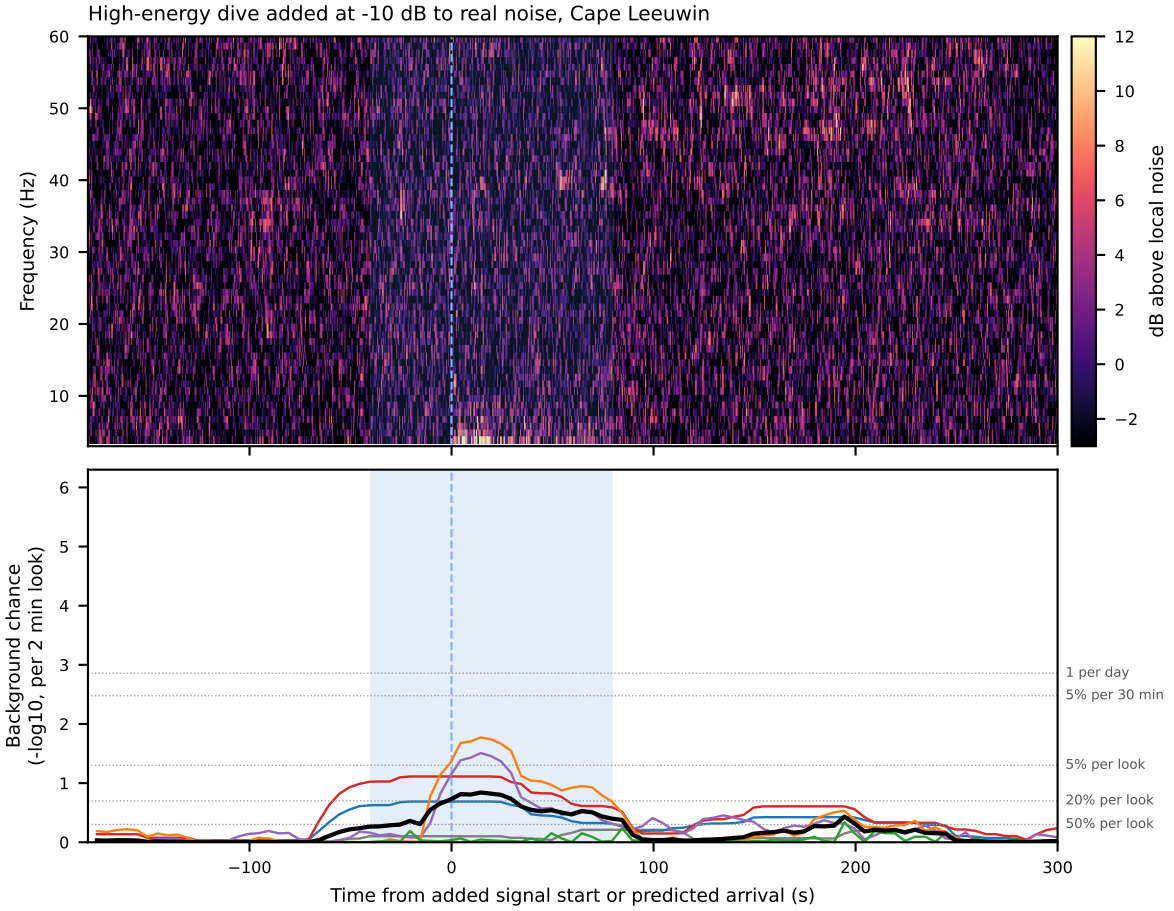
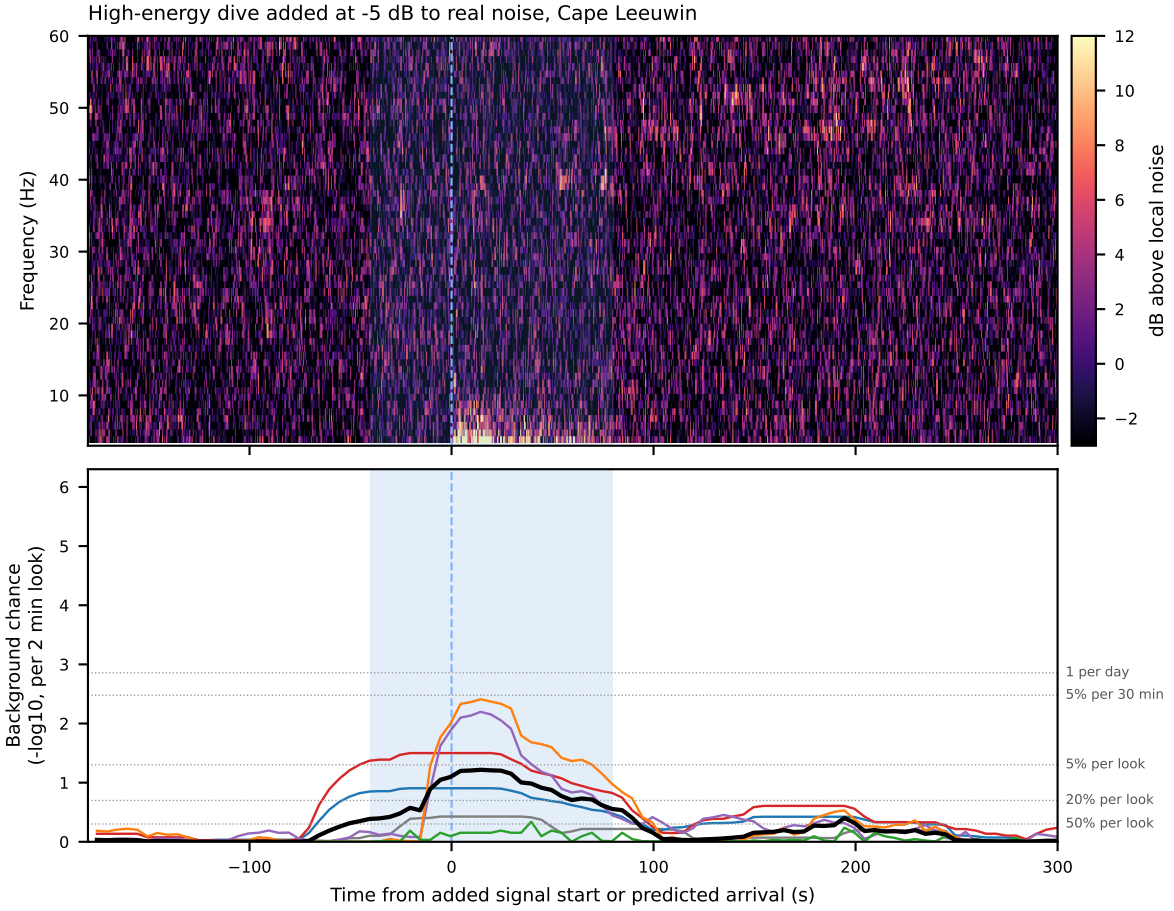


Lowest background chance
in the shaded look window

Combined	0.14
Band energy	0.62
Impact spectral templates	0.2
Spectral shape change	0.077
Correlation, no template	0.031
Correlation, impact pattern	0.017
Envelope persistence	0.65

Case 0; host record H01W1_rnd158

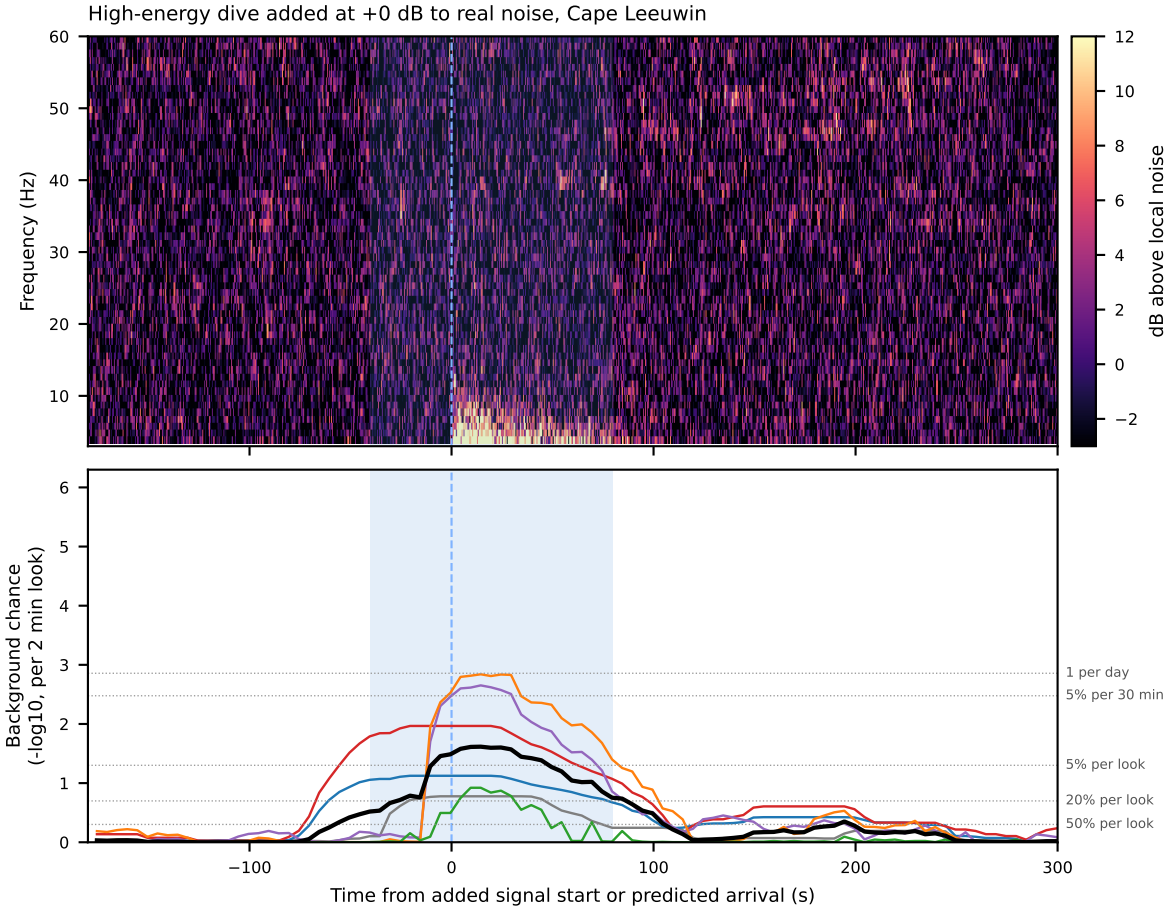


Lowest background chance
in the shaded look window

Combined	0.061
Band energy	0.37
Impact spectral templates	0.12
Spectral shape change	0.032
Correlation, no template	0.0063
Correlation, impact pattern	0.0039
Envelope persistence	0.46

Case 0; host record H01W1_rnd158

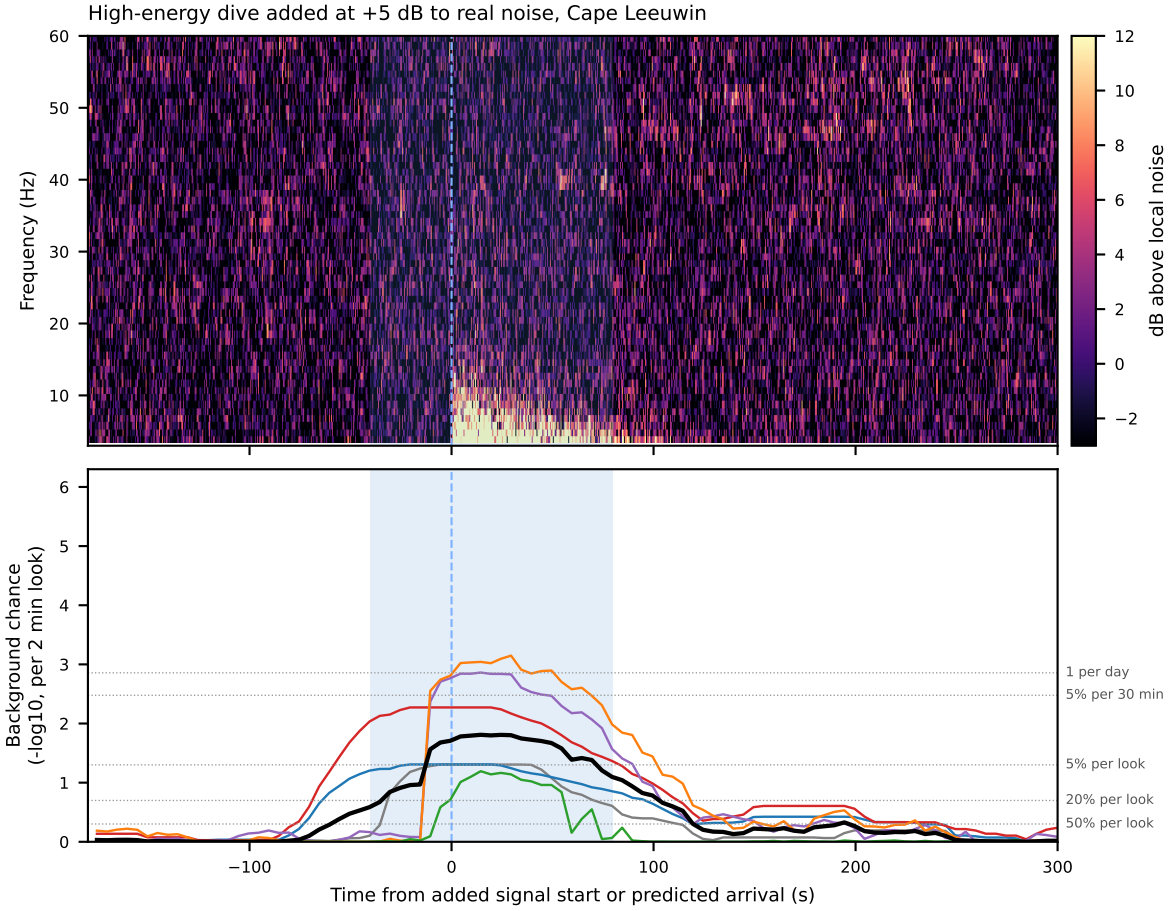
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.024
Band energy	0.17
Impact spectral templates	0.075
Spectral shape change	0.011
Correlation, no template	0.0022
Correlation, impact pattern	0.0014
Envelope persistence	0.12

Case 0; host record H01W1_rnd158

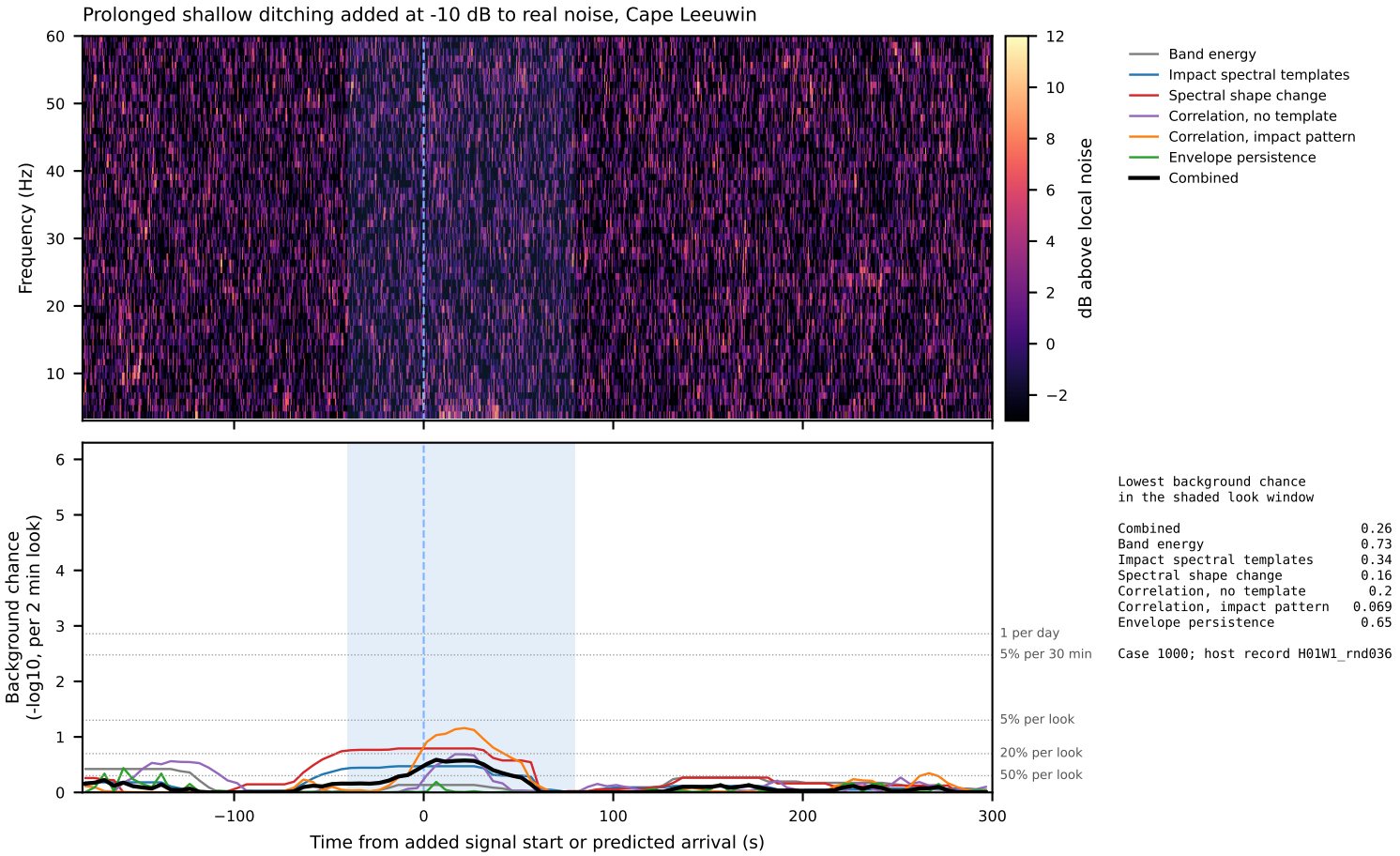


Lowest background chance
in the shaded look window

Combined	0.016
Band energy	0.049
Impact spectral templates	0.049
Spectral shape change	0.0053
Correlation, no template	0.0014
Correlation, impact pattern	0.00071
Envelope persistence	0.064

Case 0; host record H01W1_rnd158

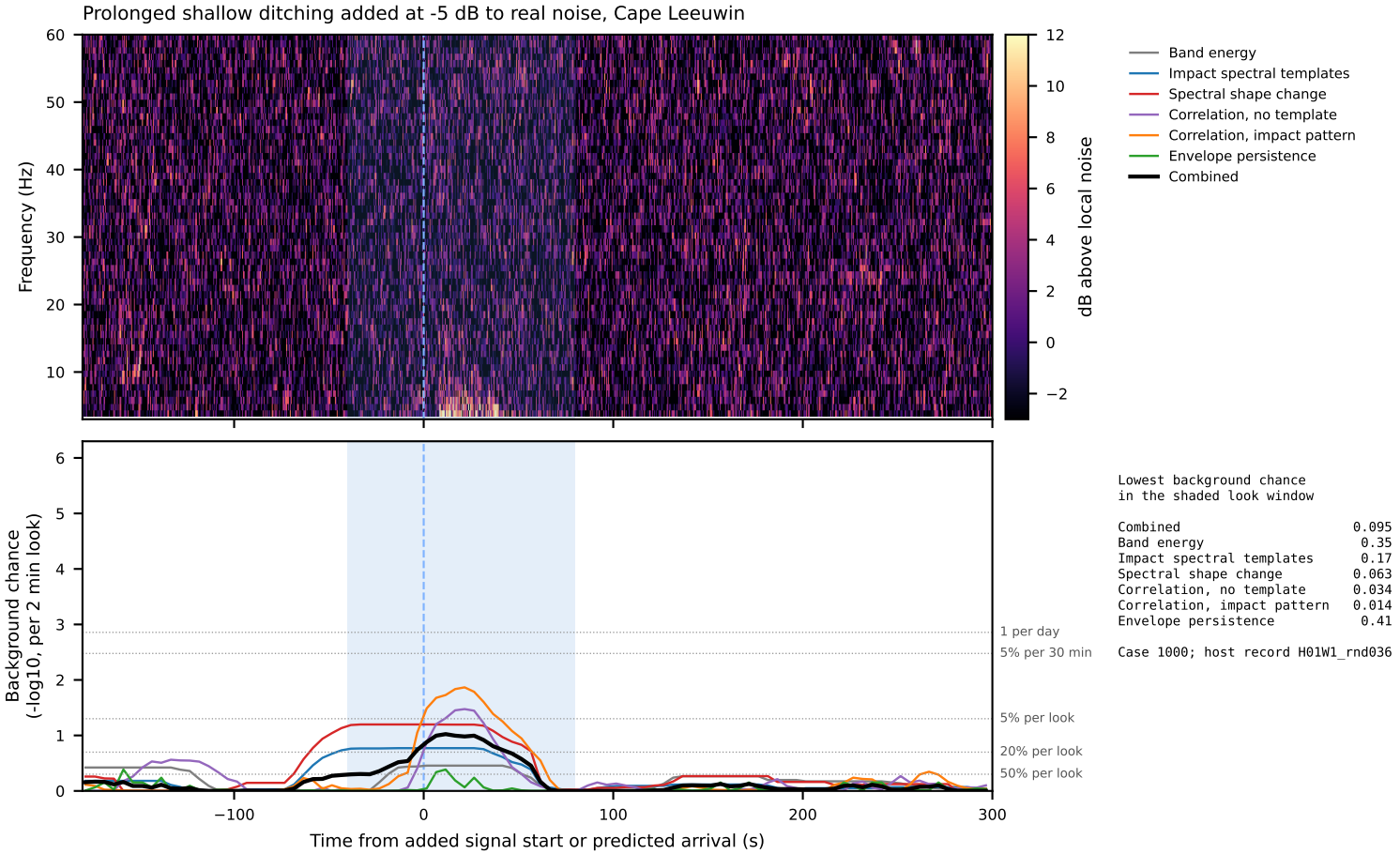
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.26
Band energy	0.73
Impact spectral templates	0.34
Spectral shape change	0.16
Correlation, no template	0.2
Correlation, impact pattern	0.069
Envelope persistence	0.65

Case 1000; host record H01W1_rnd036

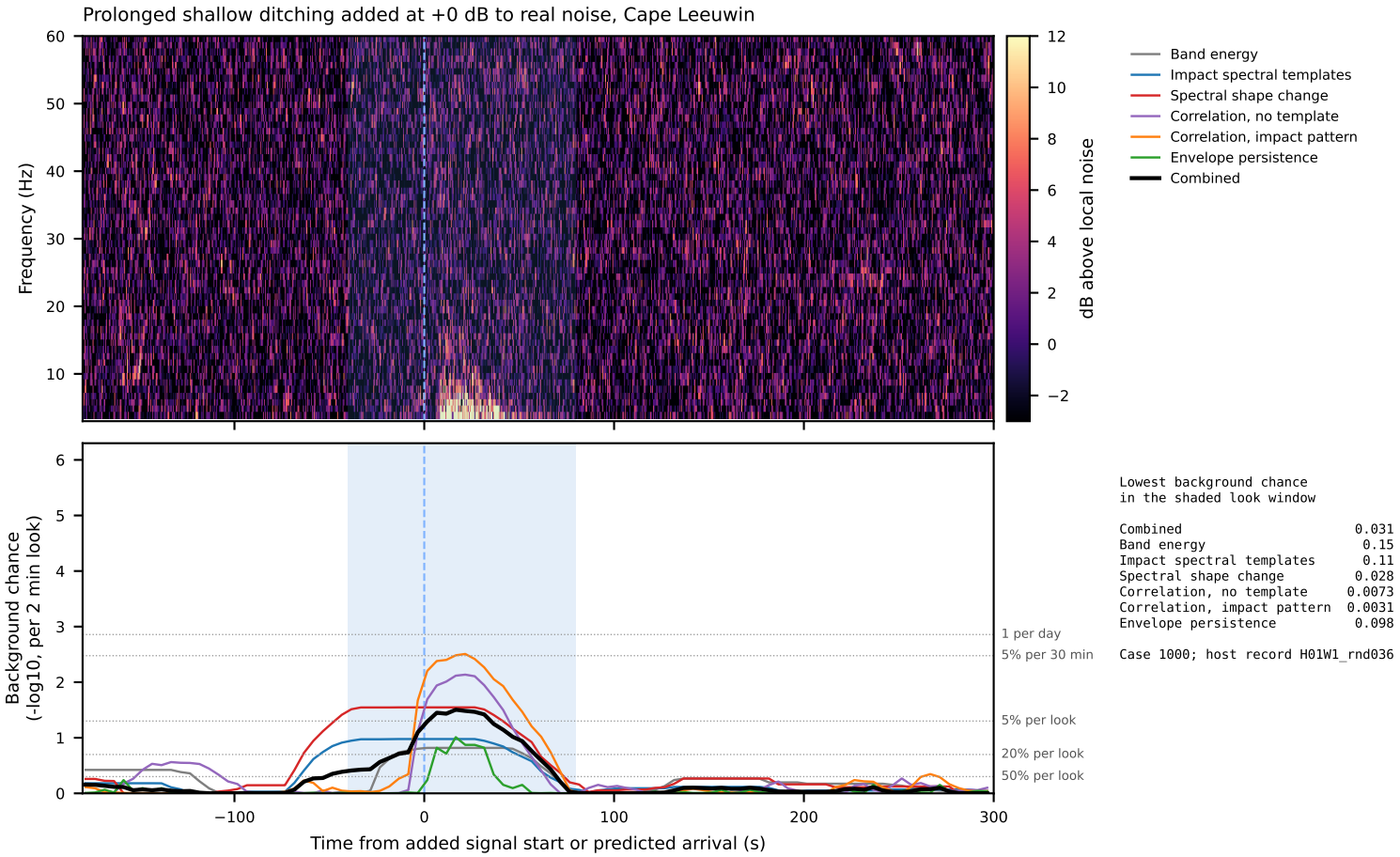


Lowest background chance
in the shaded look window

Combined	0.095
Band energy	0.35
Impact spectral templates	0.17
Spectral shape change	0.063
Correlation, no template	0.034
Correlation, impact pattern	0.014
Envelope persistence	0.41

Case 1000; host record H01W1_rnd036

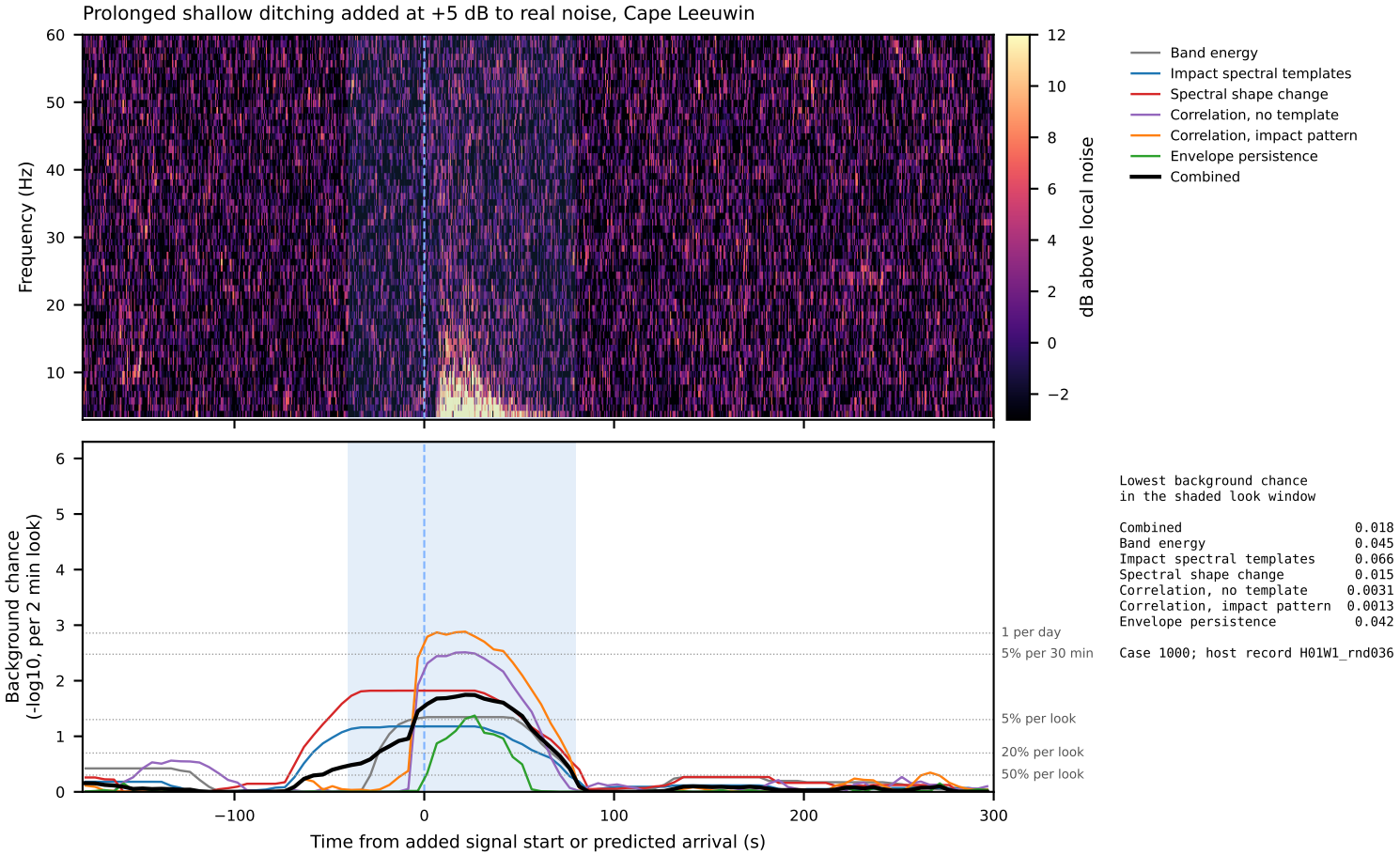
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.031
Band energy	0.15
Impact spectral templates	0.11
Spectral shape change	0.028
Correlation, no template	0.0073
Correlation, impact pattern	0.0031
Envelope persistence	0.098

Case 1000; host record H01W1_rnd036

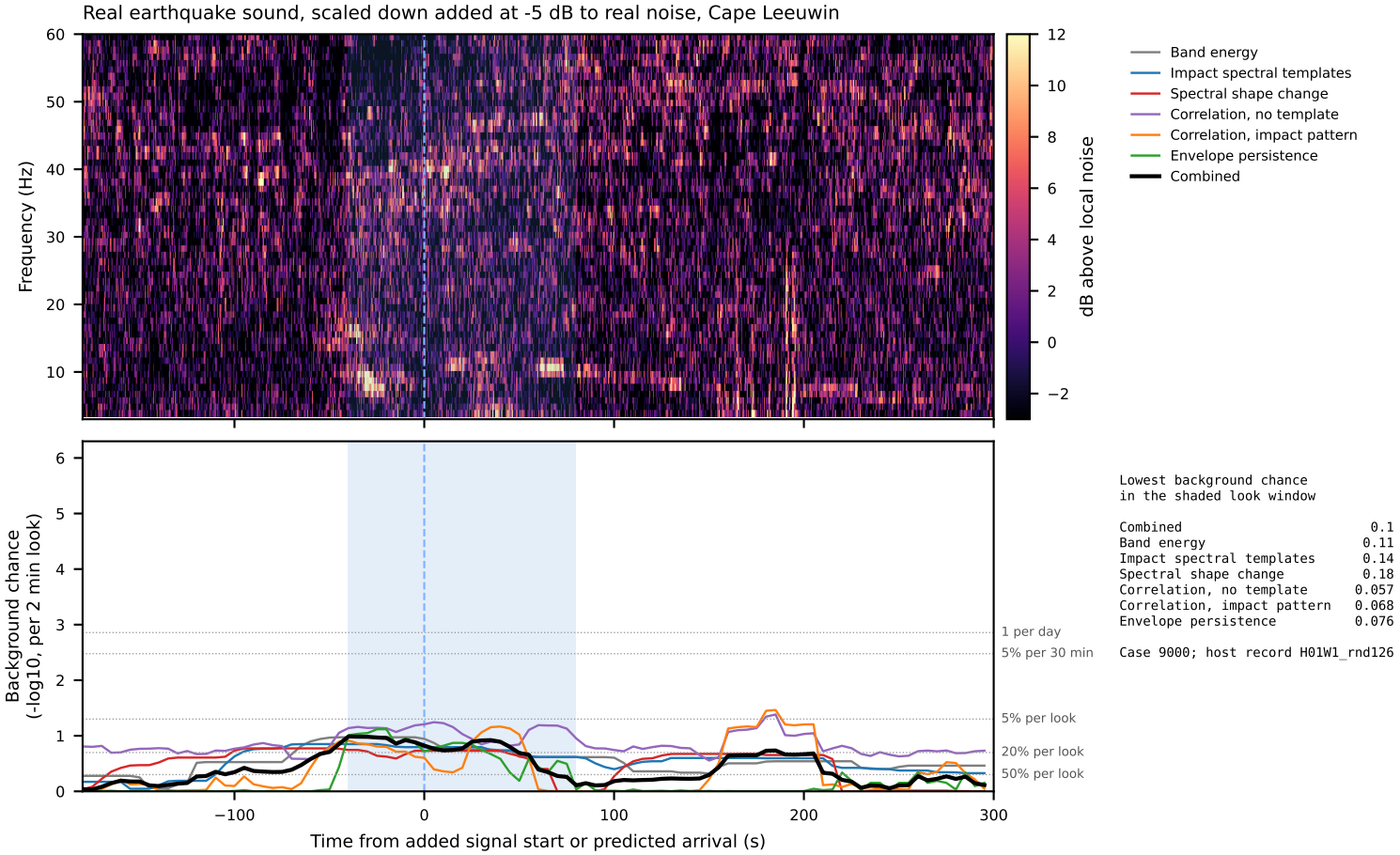


Lowest background chance
in the shaded look window

Combined	0.018
Band energy	0.045
Impact spectral templates	0.066
Spectral shape change	0.015
Correlation, no template	0.0031
Correlation, impact pattern	0.0013
Envelope persistence	0.042

Case 1000; host record H01W1_rnd036

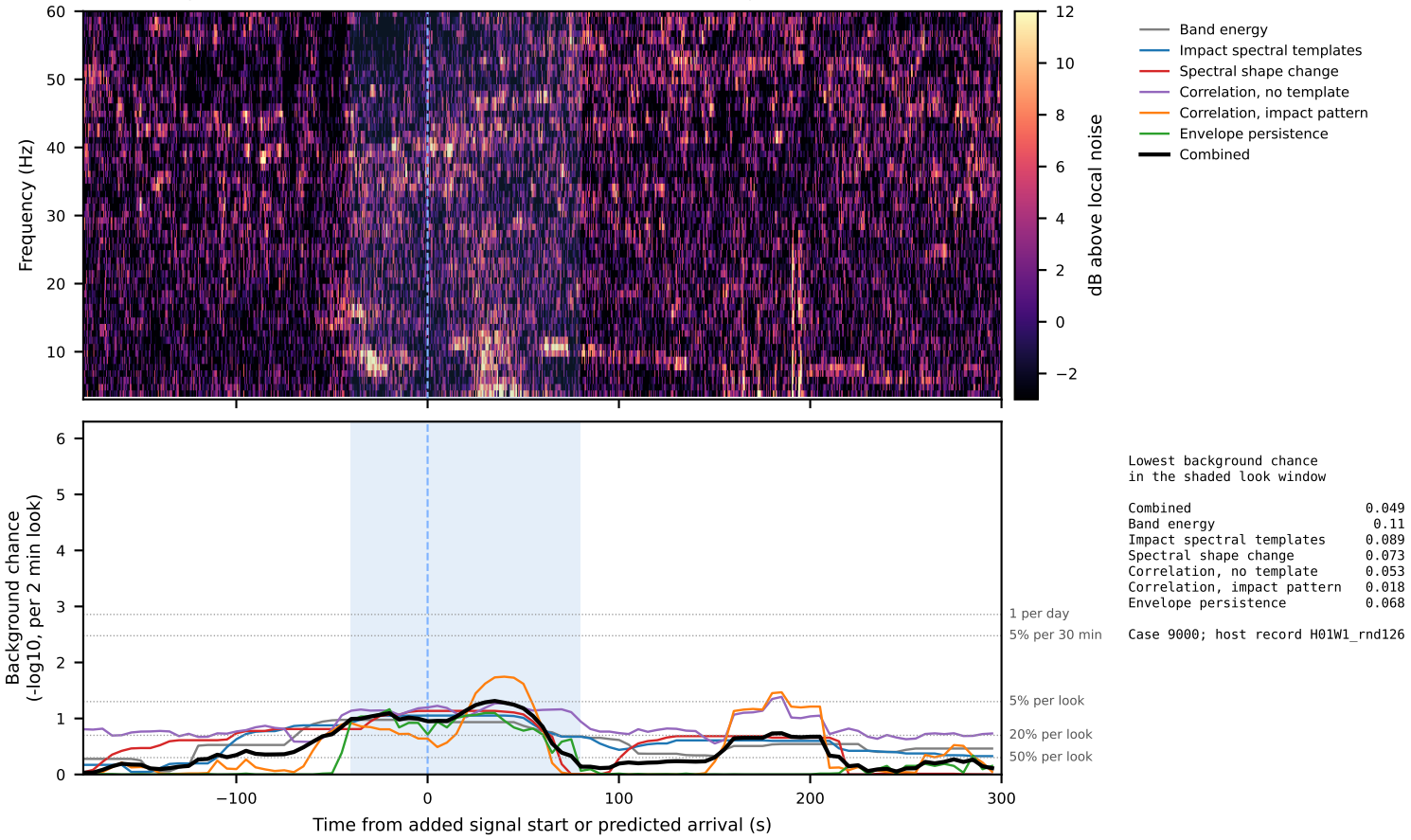
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.1
Band energy	0.11
Impact spectral templates	0.14
Spectral shape change	0.18
Correlation, no template	0.057
Correlation, impact pattern	0.068
Envelope persistence	0.076

Case 9000; host record H01W1_rnd126

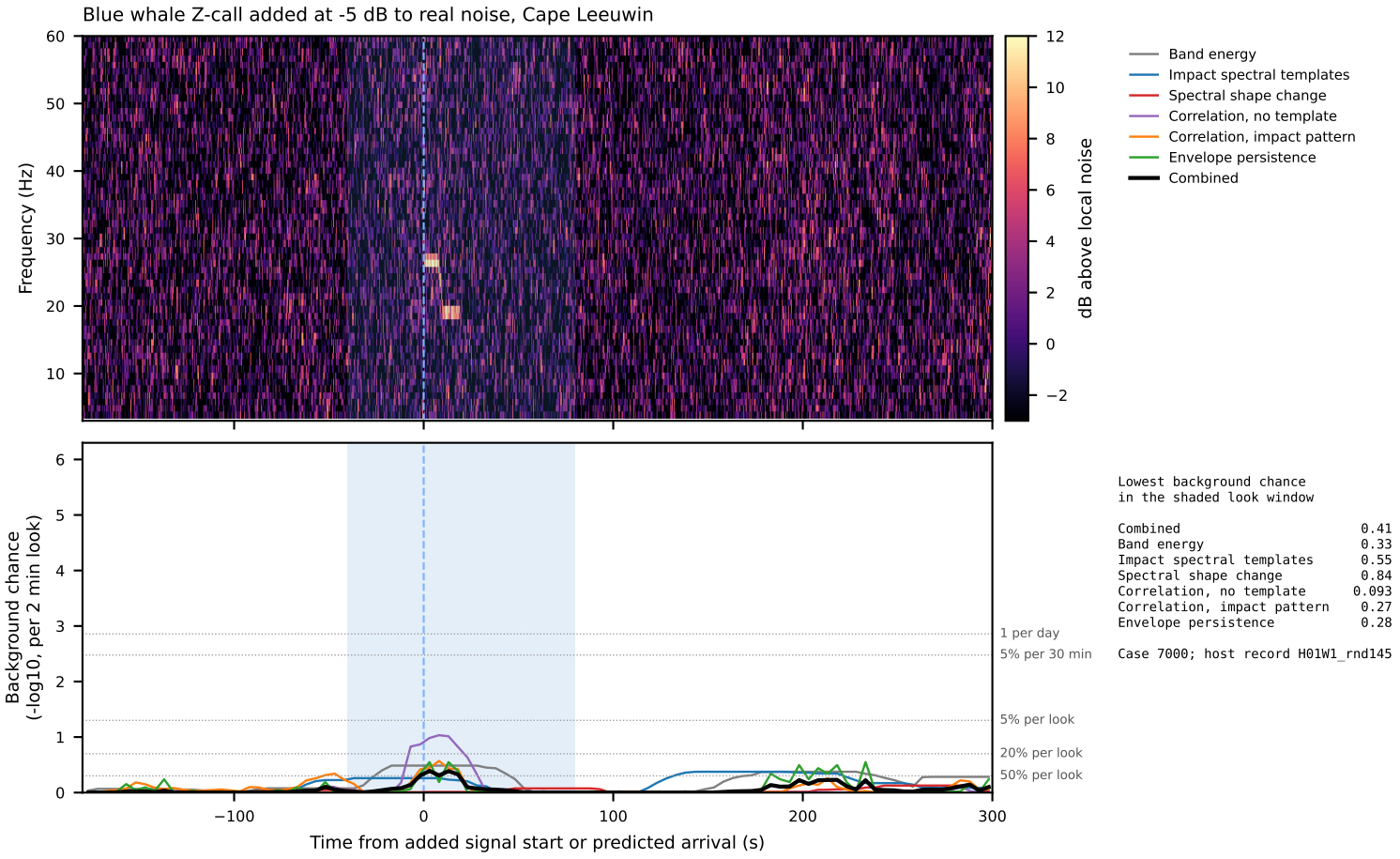


Lowest background chance
in the shaded look window

Combined	0.049
Band energy	0.11
Impact spectral templates	0.089
Spectral shape change	0.073
Correlation, no template	0.053
Correlation, impact pattern	0.018
Envelope persistence	0.068

Case 9000; host record H01W1_rnd126

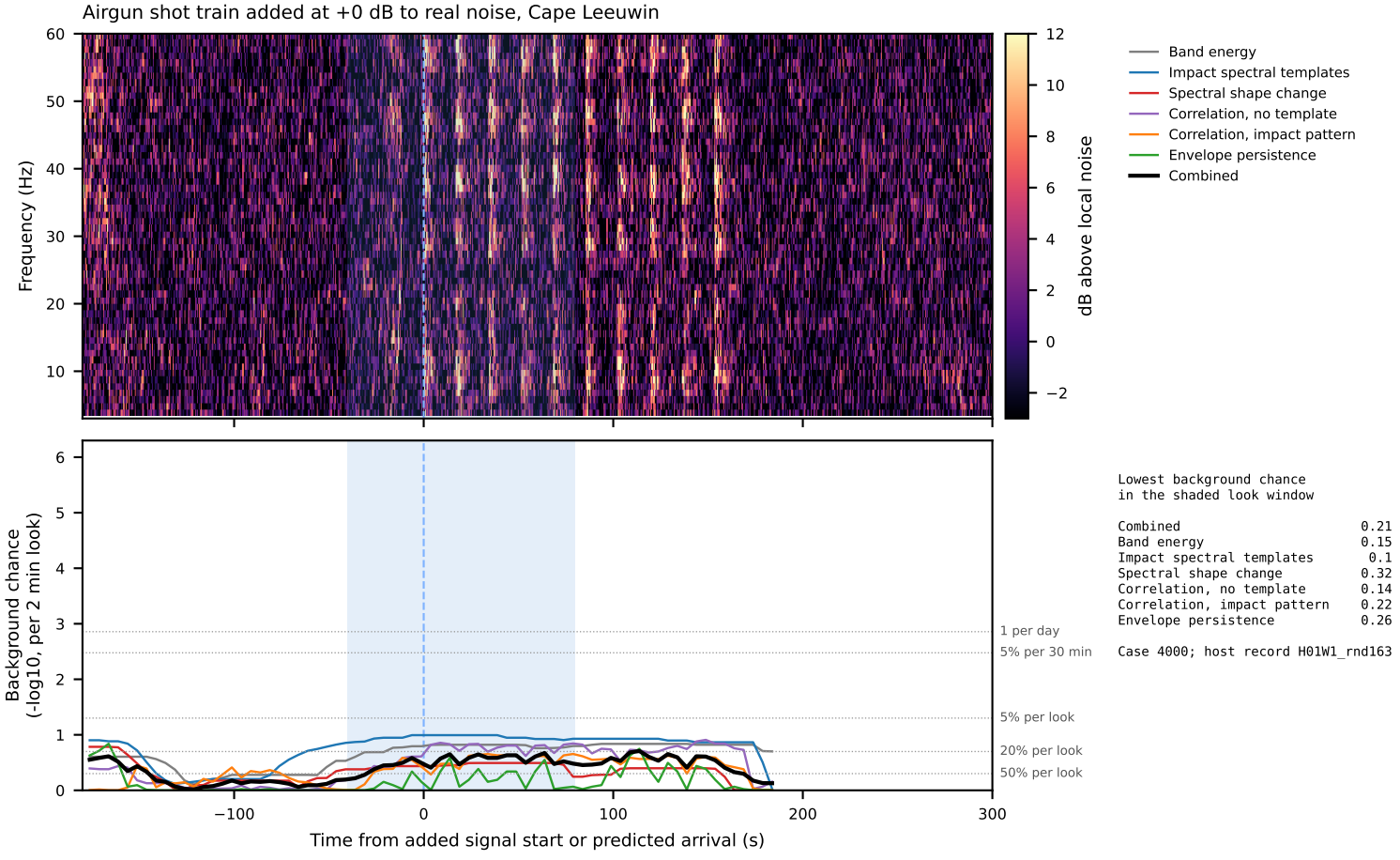
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.41
Band energy	0.33
Impact spectral templates	0.55
Spectral shape change	0.84
Correlation, no template	0.093
Correlation, impact pattern	0.27
Envelope persistence	0.28

Case 7000; host record H01W1_rnd145

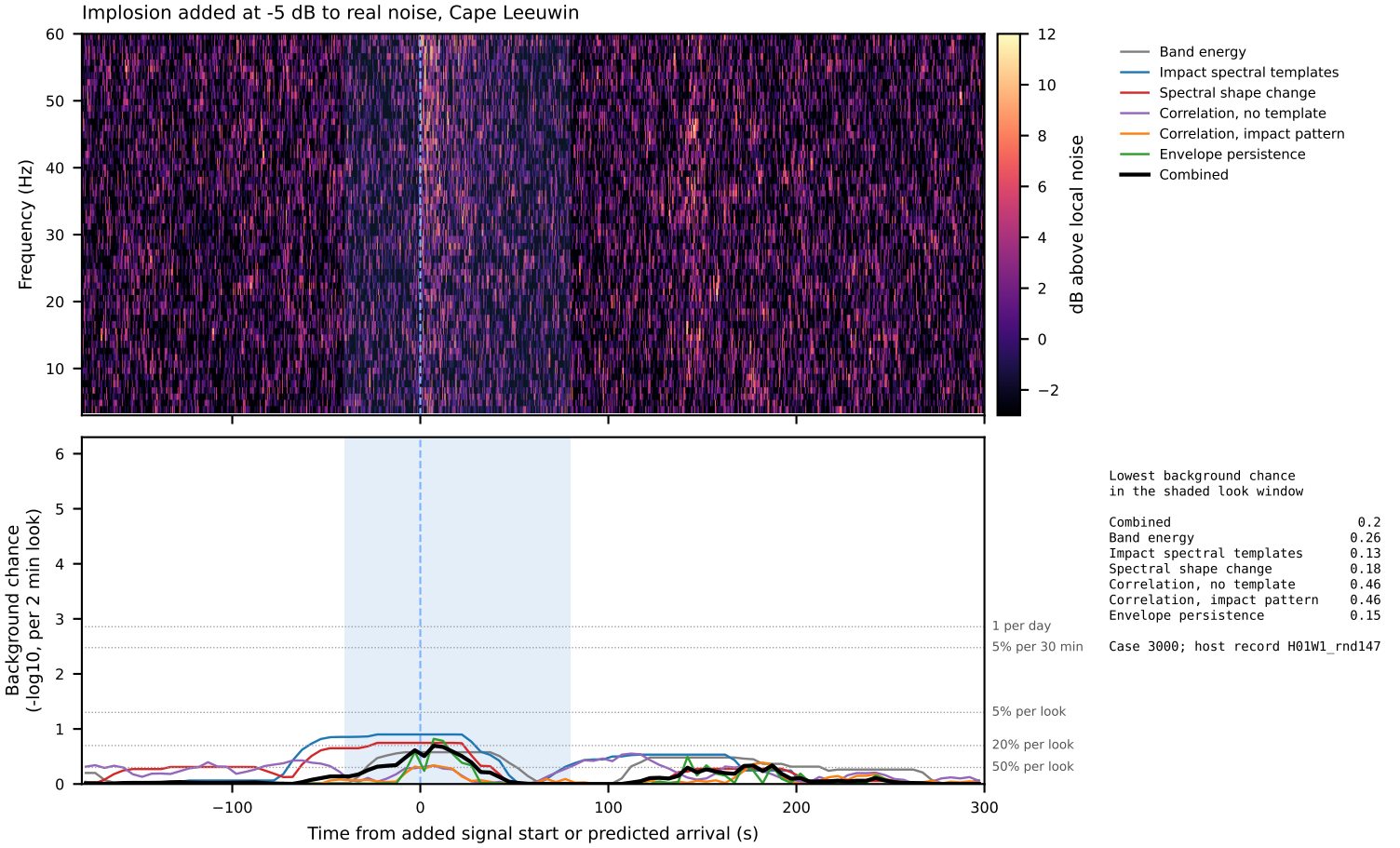


Lowest background chance
in the shaded look window

Combined	0.21
Band energy	0.15
Impact spectral templates	0.1
Spectral shape change	0.32
Correlation, no template	0.14
Correlation, impact pattern	0.22
Envelope persistence	0.26

Case 4000; host record H01W1_rnd163

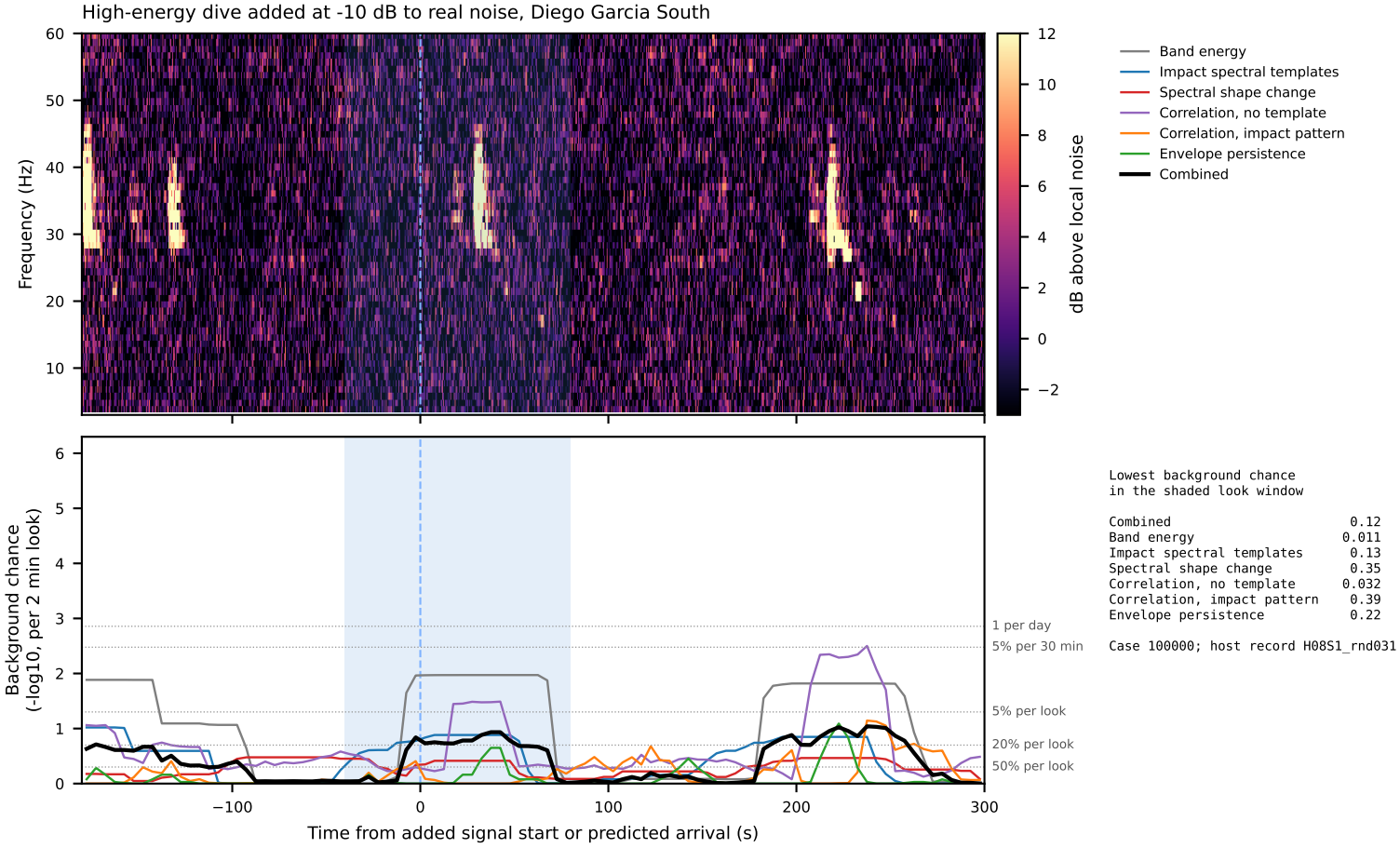
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.2
Band energy	0.26
Impact spectral templates	0.13
Spectral shape change	0.18
Correlation, no template	0.46
Correlation, impact pattern	0.46
Envelope persistence	0.15

Case 3000; host record H01W1_rnd147

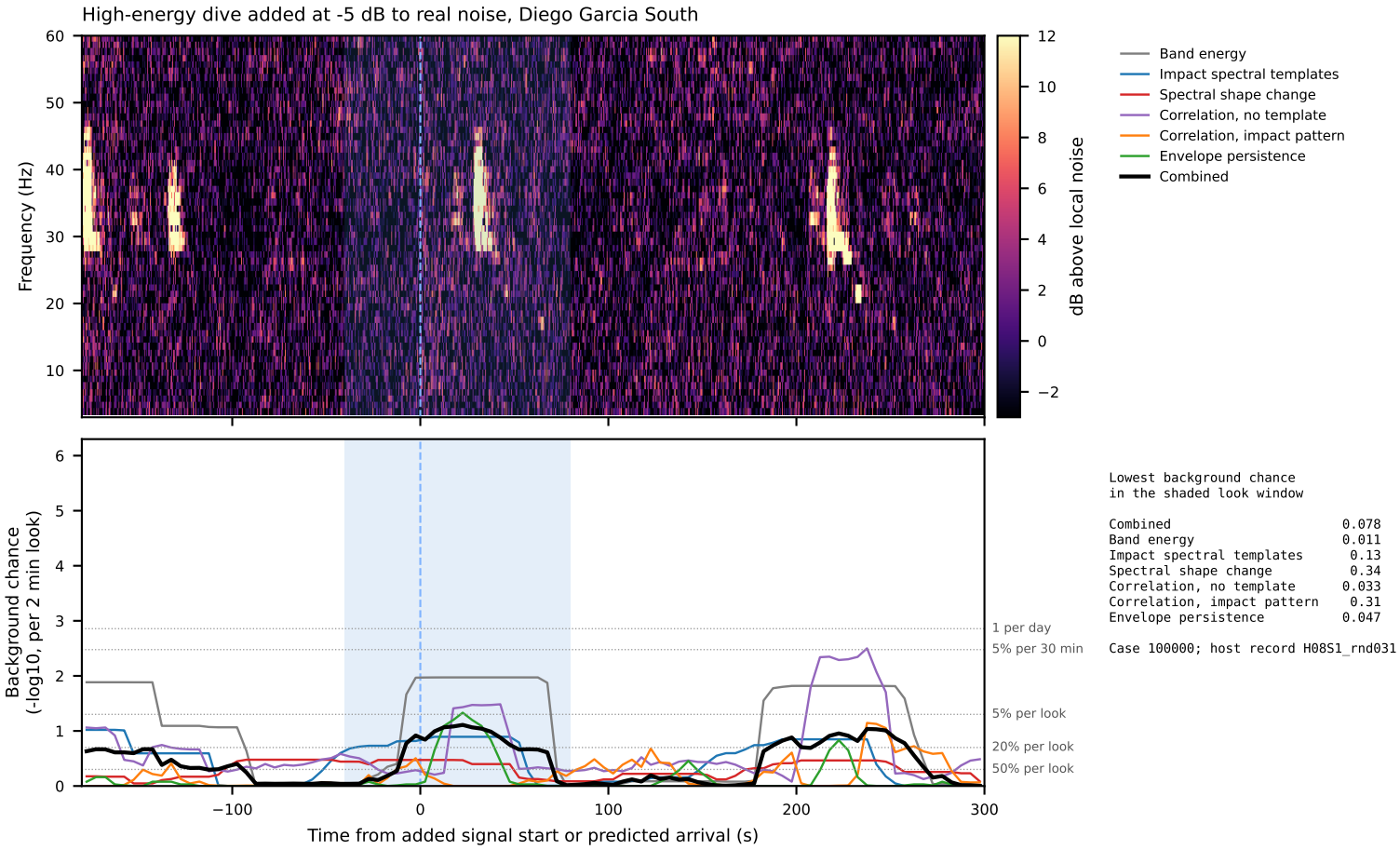


Lowest background chance
in the shaded look window

Combined	0.12
Band energy	0.011
Impact spectral templates	0.13
Spectral shape change	0.35
Correlation, no template	0.032
Correlation, impact pattern	0.39
Envelope persistence	0.22

Case 100000; host record H08S1_rnd031

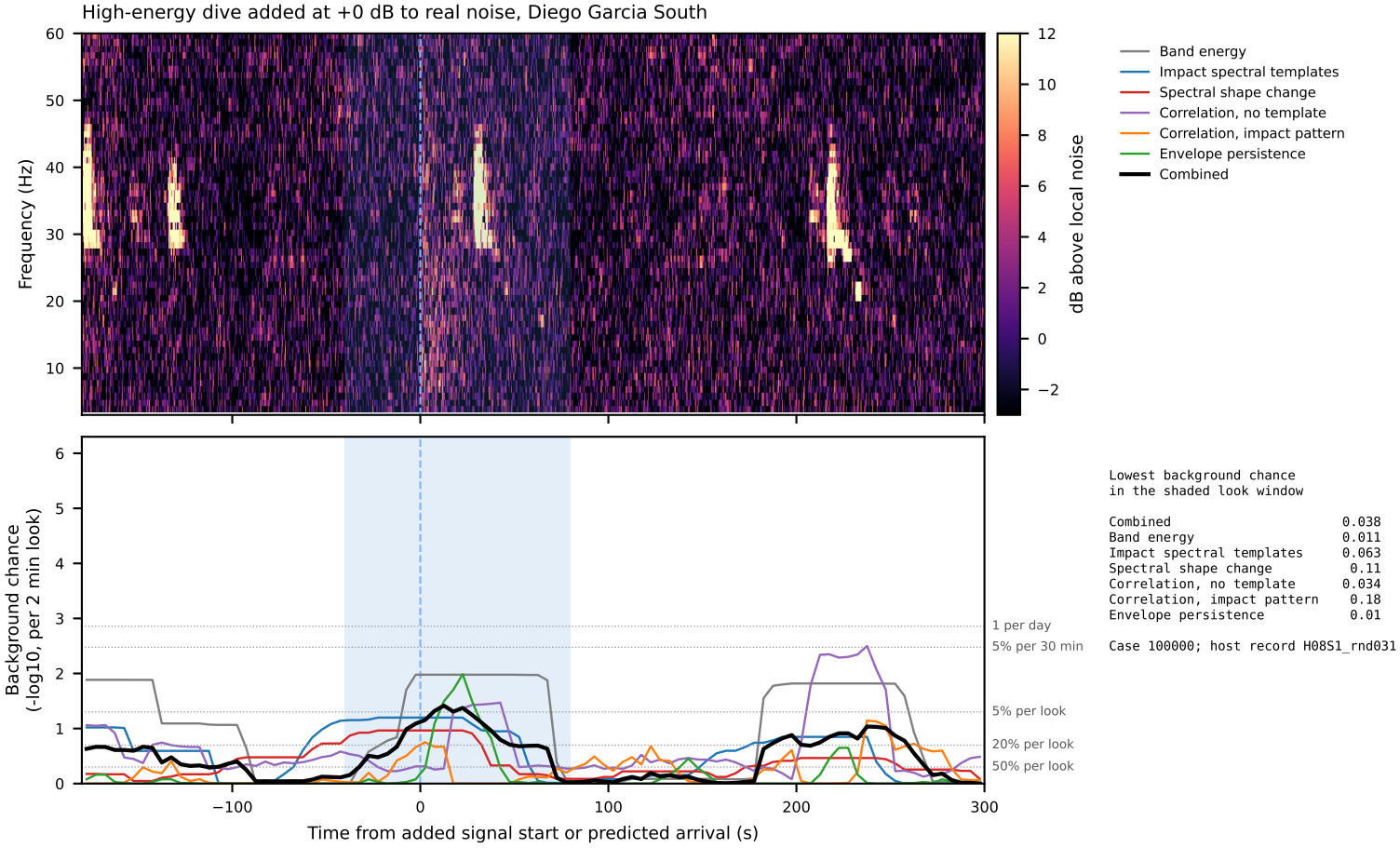
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.078
Band energy	0.011
Impact spectral templates	0.13
Spectral shape change	0.34
Correlation, no template	0.033
Correlation, impact pattern	0.31
Envelope persistence	0.047

Case 100000; host record H08S1_rnd031

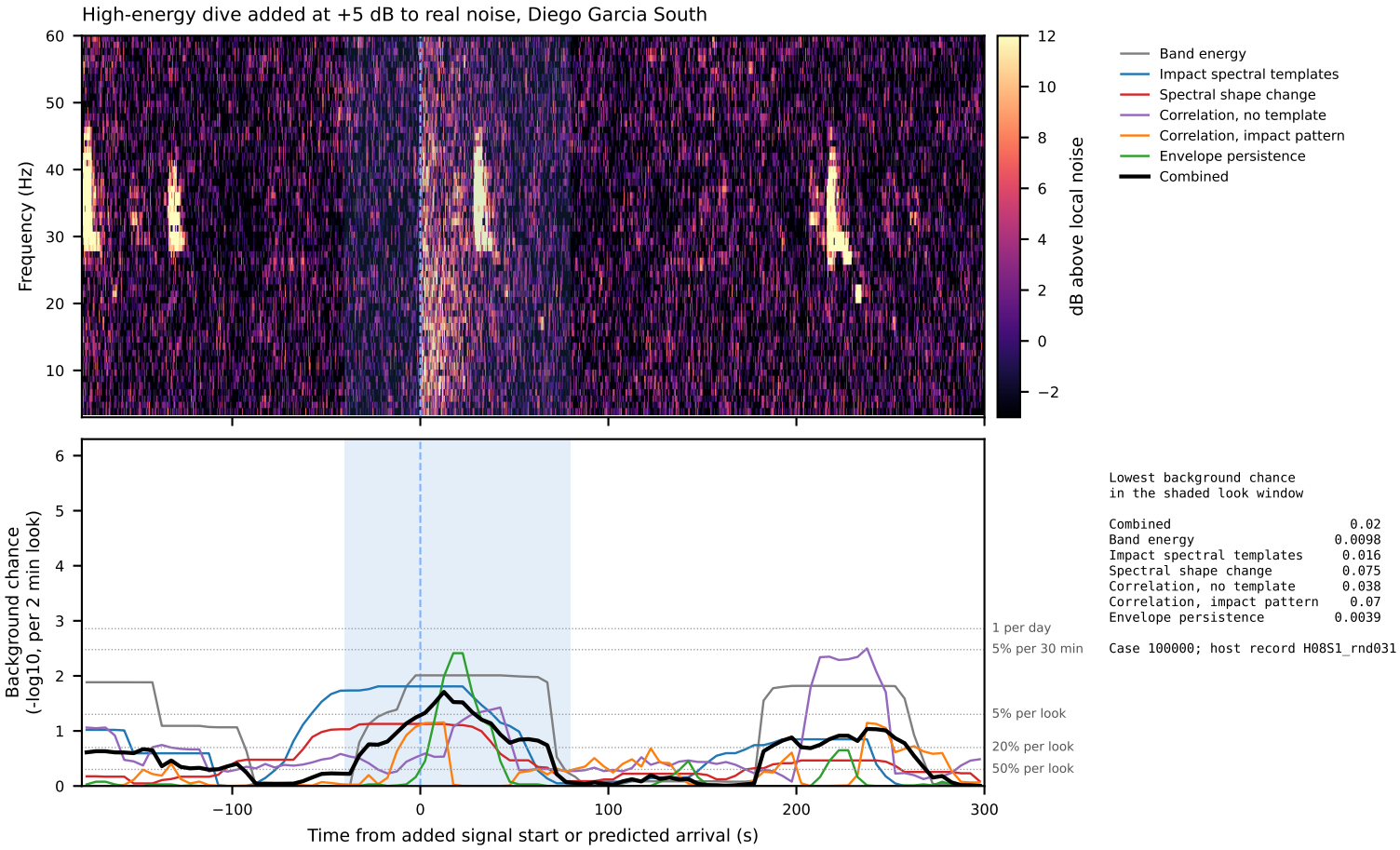


Lowest background chance
in the shaded look window

Combined	0.038
Band energy	0.011
Impact spectral templates	0.063
Spectral shape change	0.11
Correlation, no template	0.034
Correlation, impact pattern	0.18
Envelope persistence	0.01

Case 100000; host record H08S1_rnd031

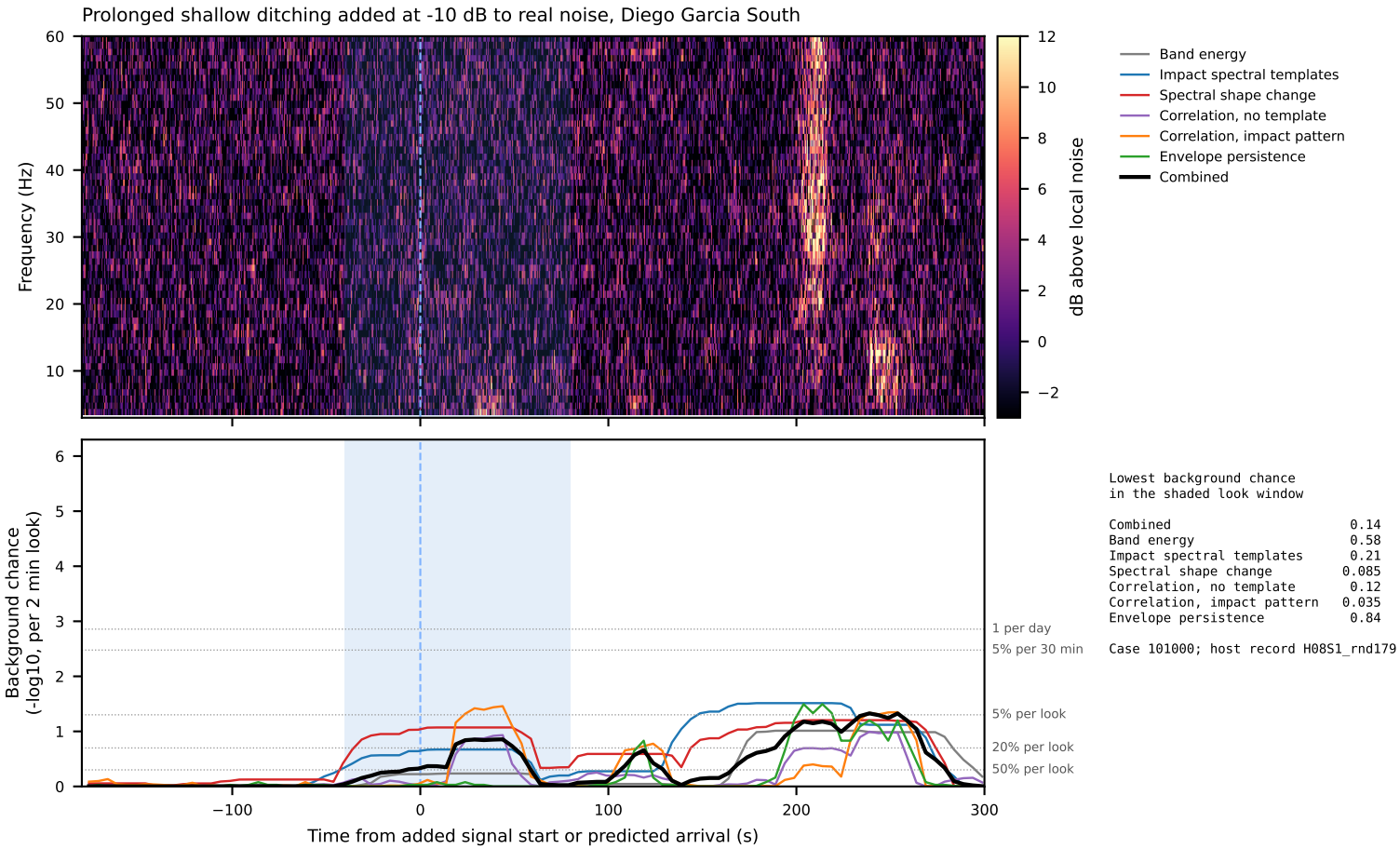
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.02
Band energy	0.0098
Impact spectral templates	0.016
Spectral shape change	0.075
Correlation, no template	0.038
Correlation, impact pattern	0.07
Envelope persistence	0.0039

Case 100000; host record H08S1_rnd031

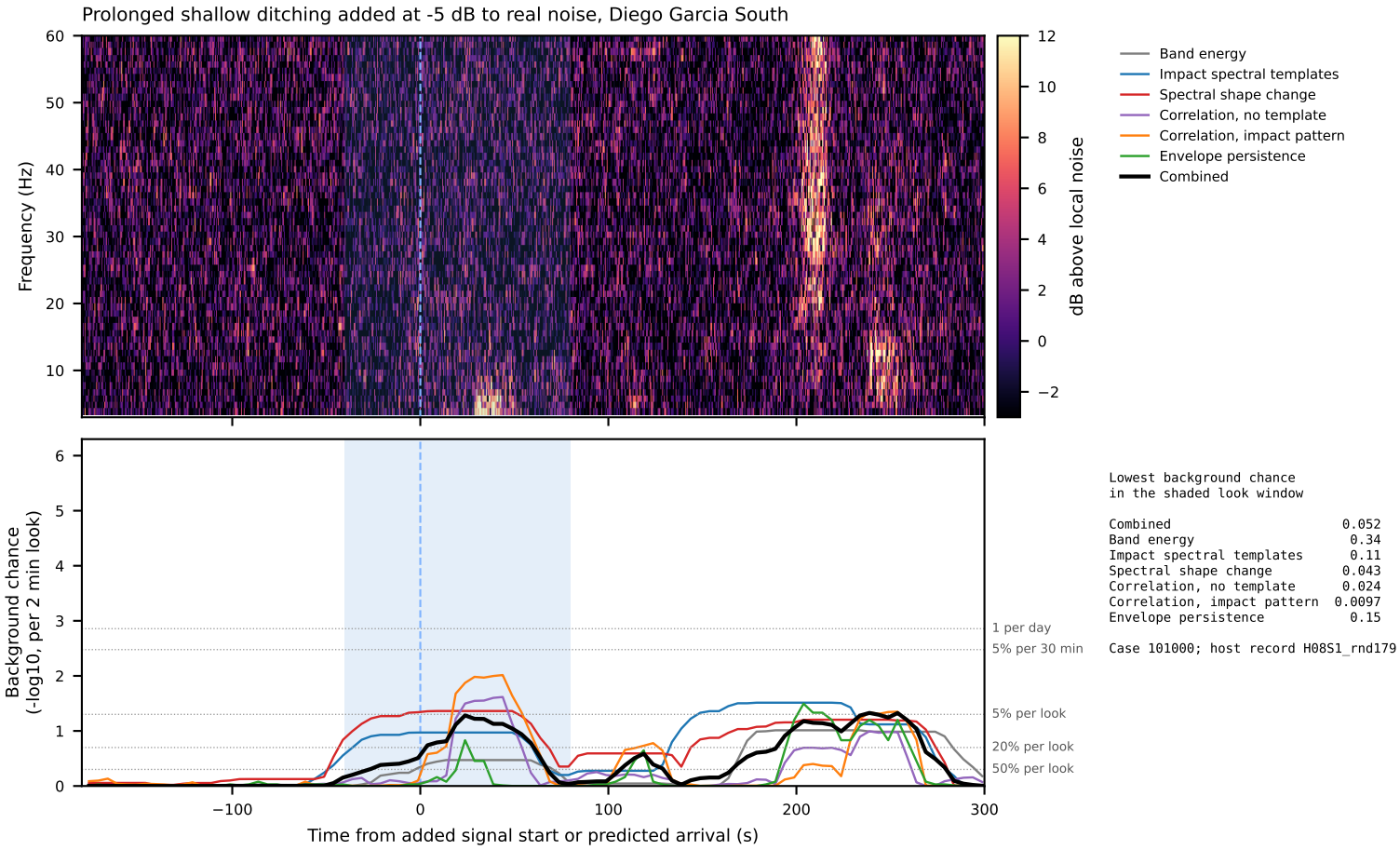


Lowest background chance
in the shaded look window

Combined	0.14
Band energy	0.58
Impact spectral templates	0.21
Spectral shape change	0.085
Correlation, no template	0.12
Correlation, impact pattern	0.035
Envelope persistence	0.84

Case 101000; host record H08S1_rnd179

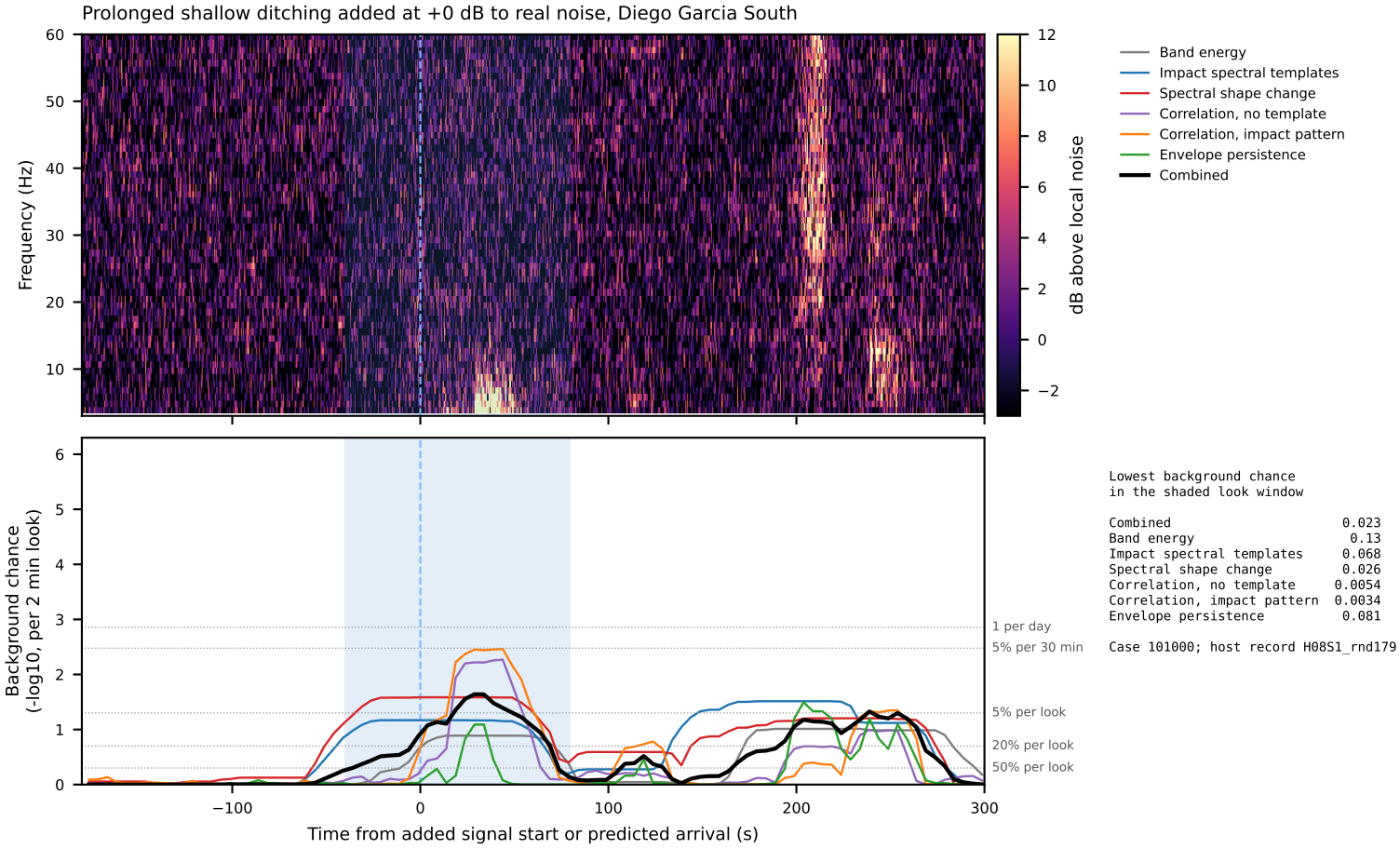
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.052
Band energy	0.34
Impact spectral templates	0.11
Spectral shape change	0.043
Correlation, no template	0.024
Correlation, impact pattern	0.0097
Envelope persistence	0.15

Case 101000; host record H08S1_rnd179

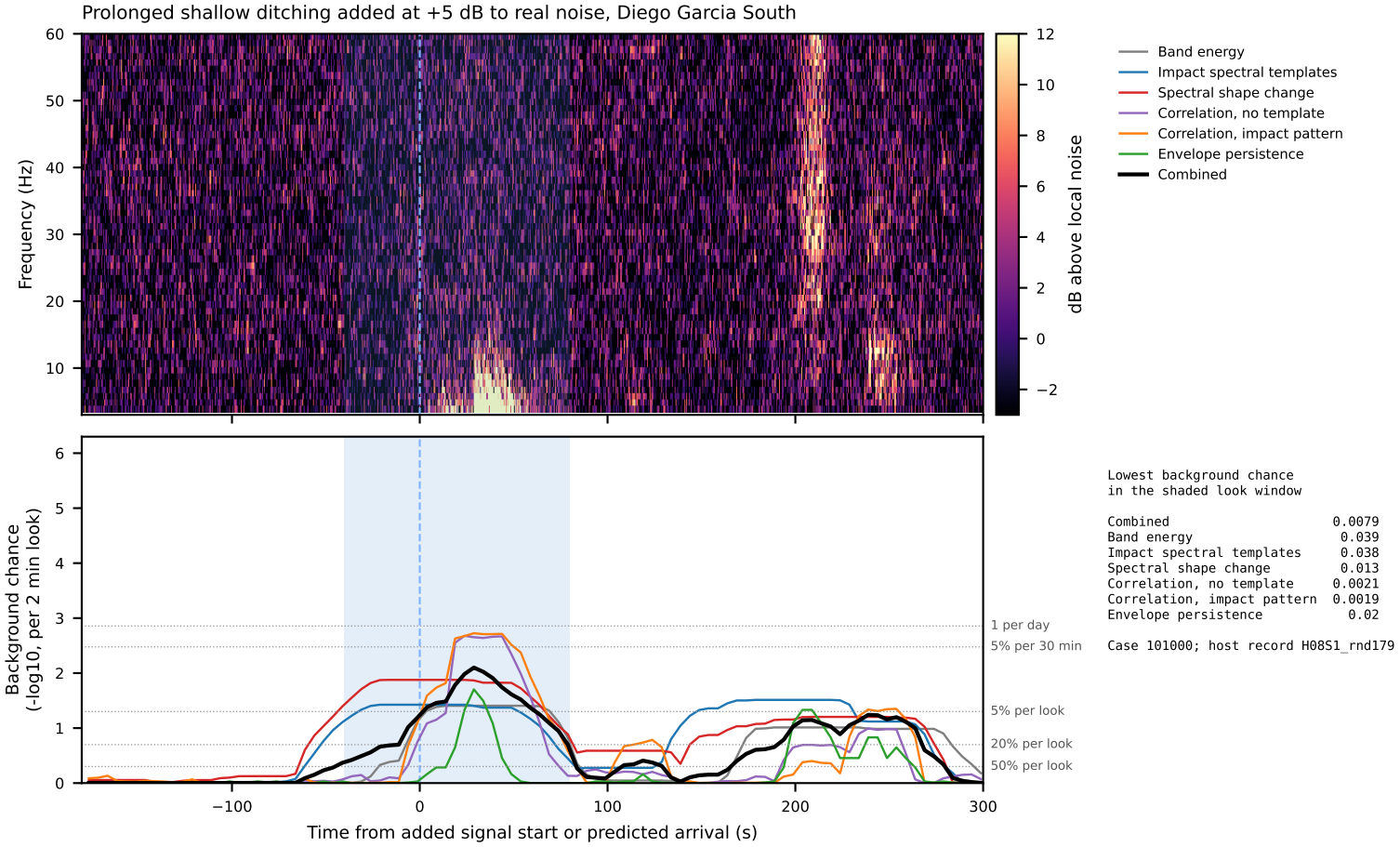


Lowest background chance
in the shaded look window

Combined	0.023
Band energy	0.13
Impact spectral templates	0.068
Spectral shape change	0.026
Correlation, no template	0.0054
Correlation, impact pattern	0.0034
Envelope persistence	0.081

Case 101000; host record H08S1_rnd179

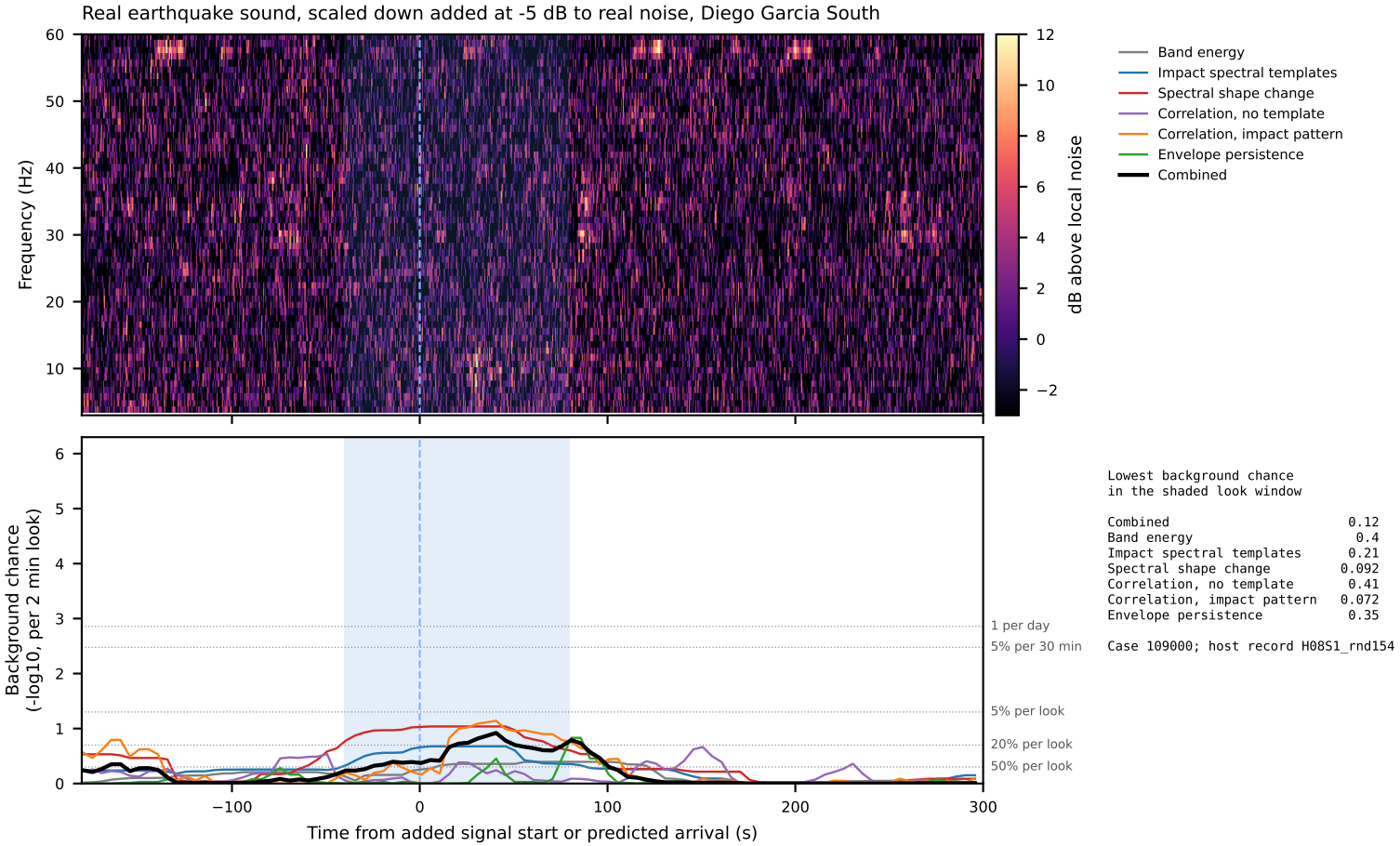
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.0079
Band energy	0.039
Impact spectral templates	0.038
Spectral shape change	0.013
Correlation, no template	0.0021
Correlation, impact pattern	0.0019
Envelope persistence	0.02

Case 101000; host record H08S1_rnd179

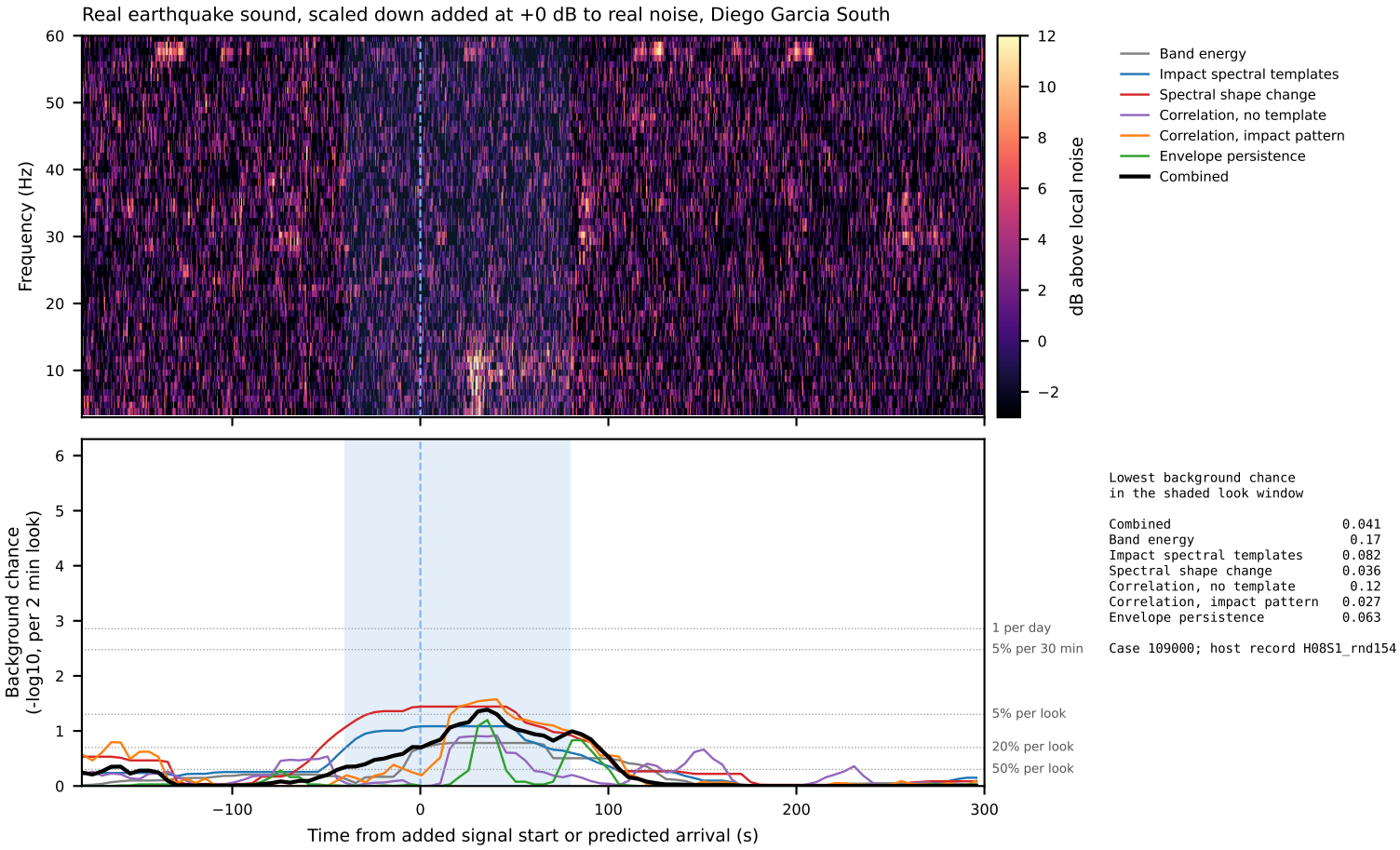


Lowest background chance
in the shaded look window

Combined	0.12
Band energy	0.4
Impact spectral templates	0.21
Spectral shape change	0.092
Correlation, no template	0.41
Correlation, impact pattern	0.072
Envelope persistence	0.35

Case 109000; host record H08S1_rnd154

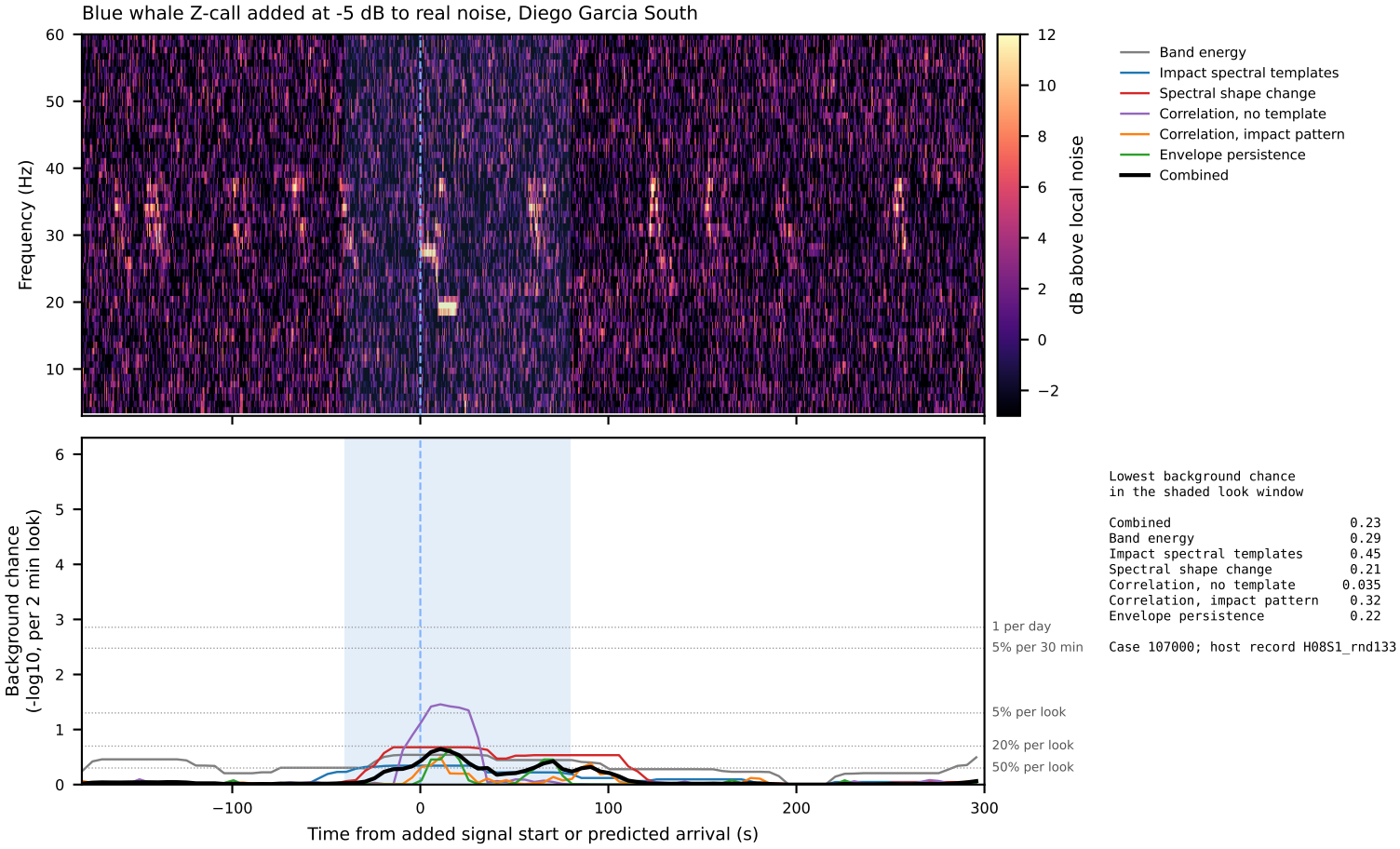
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.041
Band energy	0.17
Impact spectral templates	0.082
Spectral shape change	0.036
Correlation, no template	0.12
Correlation, impact pattern	0.027
Envelope persistence	0.063

Case 109000; host record H08S1_rnd154

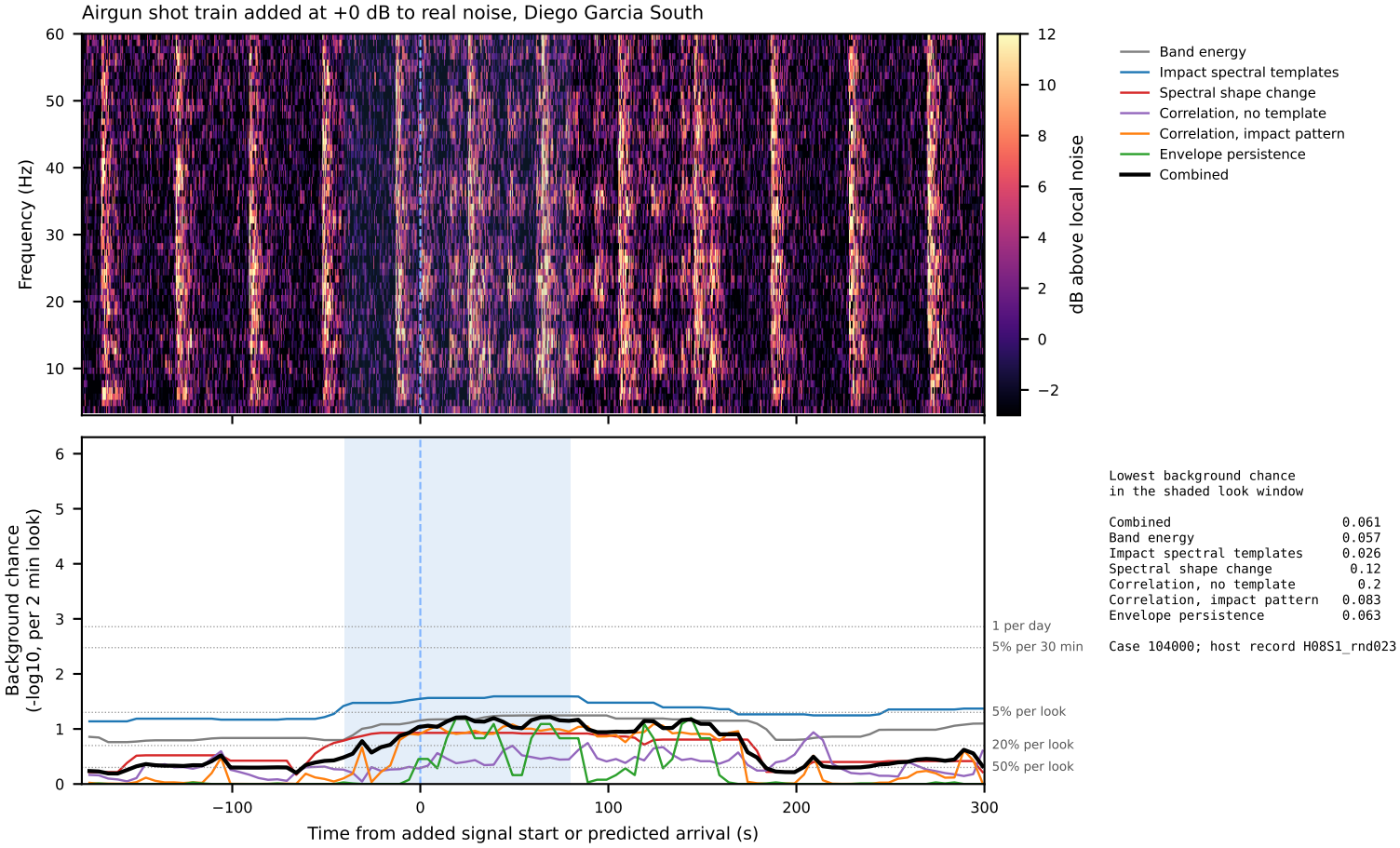


Lowest background chance
in the shaded look window

Combined	0.23
Band energy	0.29
Impact spectral templates	0.45
Spectral shape change	0.21
Correlation, no template	0.035
Correlation, impact pattern	0.32
Envelope persistence	0.22

Case 107000; host record H08S1_rnd133

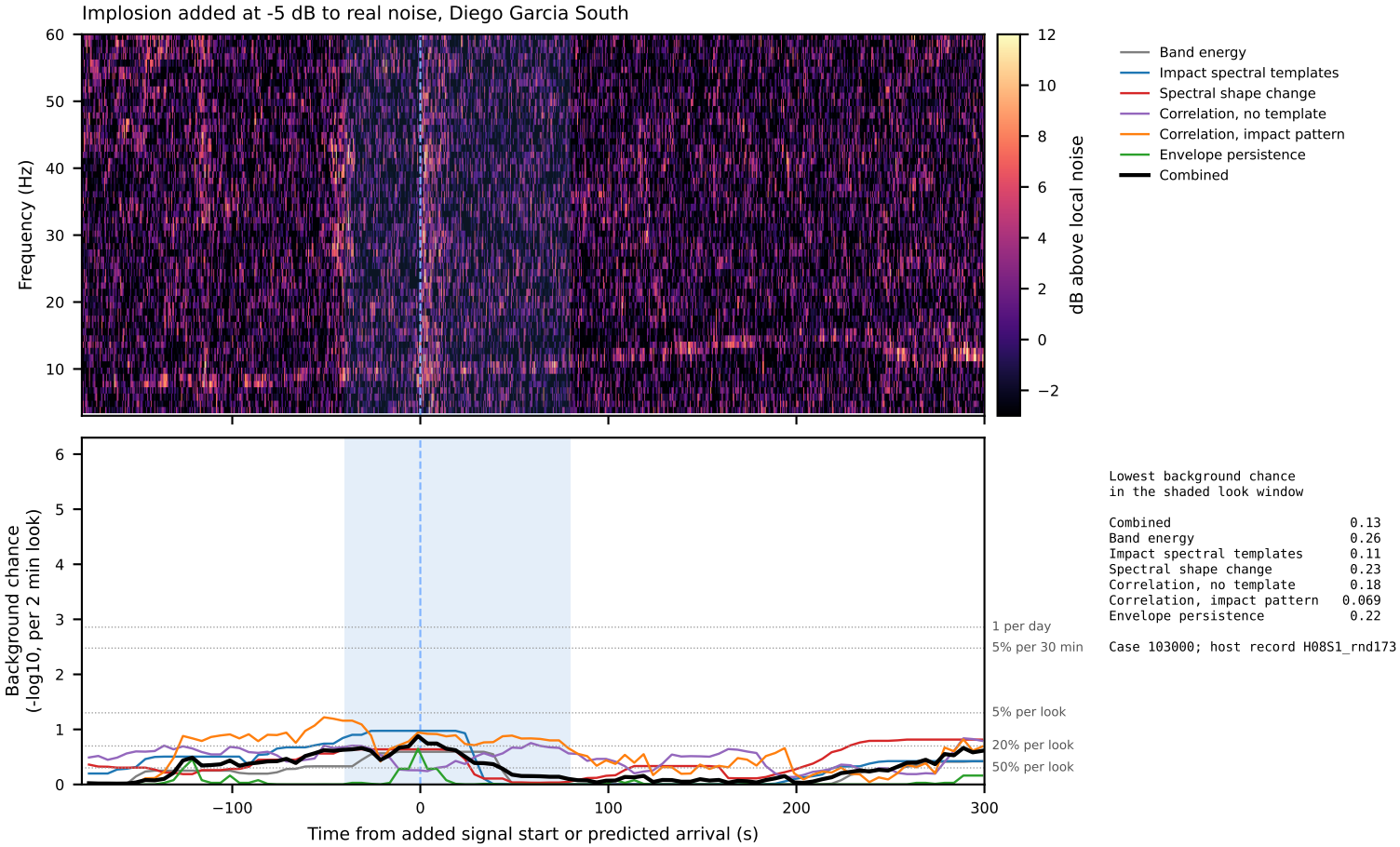
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.061
Band energy	0.057
Impact spectral templates	0.026
Spectral shape change	0.12
Correlation, no template	0.2
Correlation, impact pattern	0.083
Envelope persistence	0.063

Case 104000; host record H08S1_rnd023

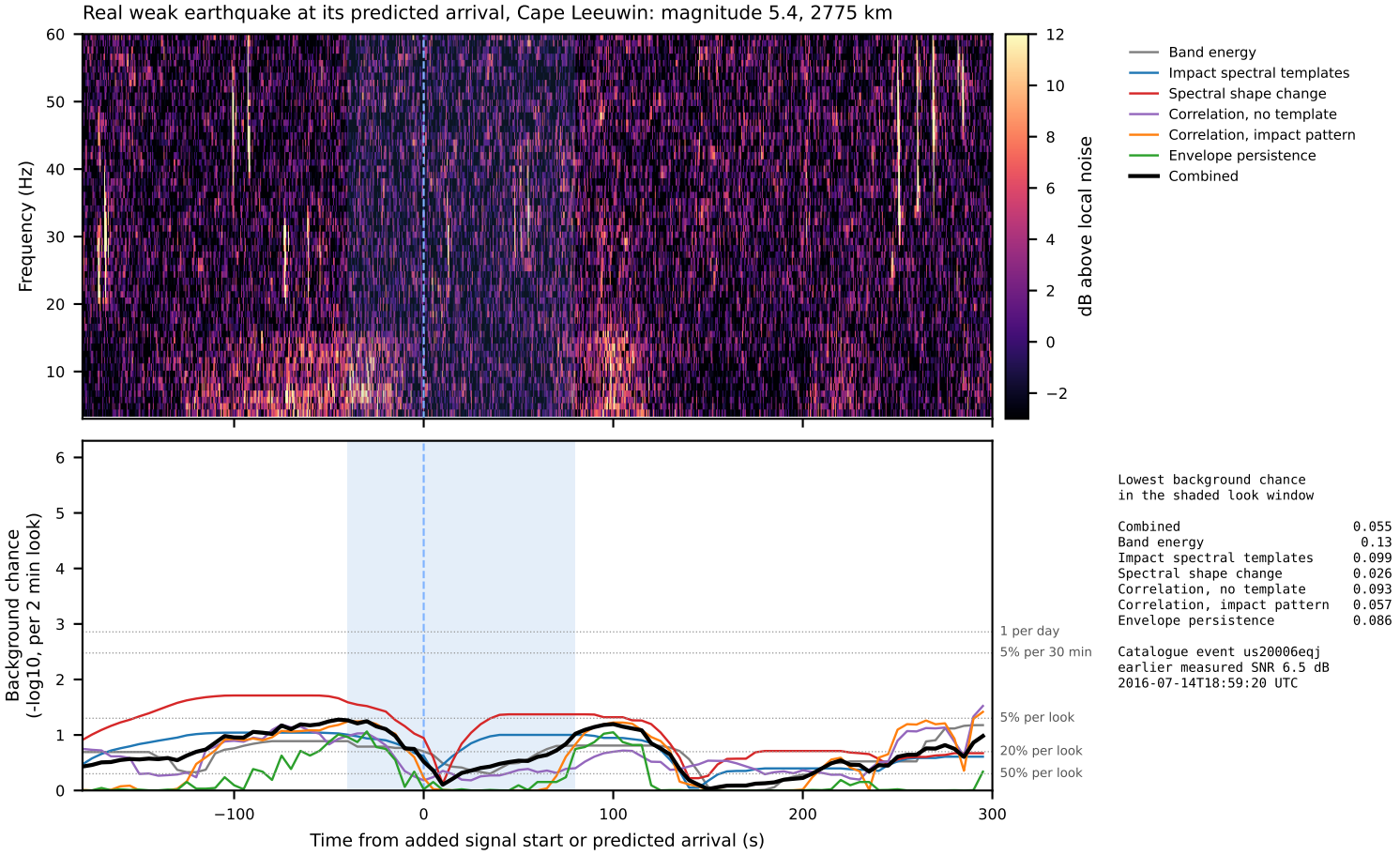


Lowest background chance
in the shaded look window

Combined	0.13
Band energy	0.26
Impact spectral templates	0.11
Spectral shape change	0.23
Correlation, no template	0.18
Correlation, impact pattern	0.069
Envelope persistence	0.22

Case 103000; host record H08S1_rnd173

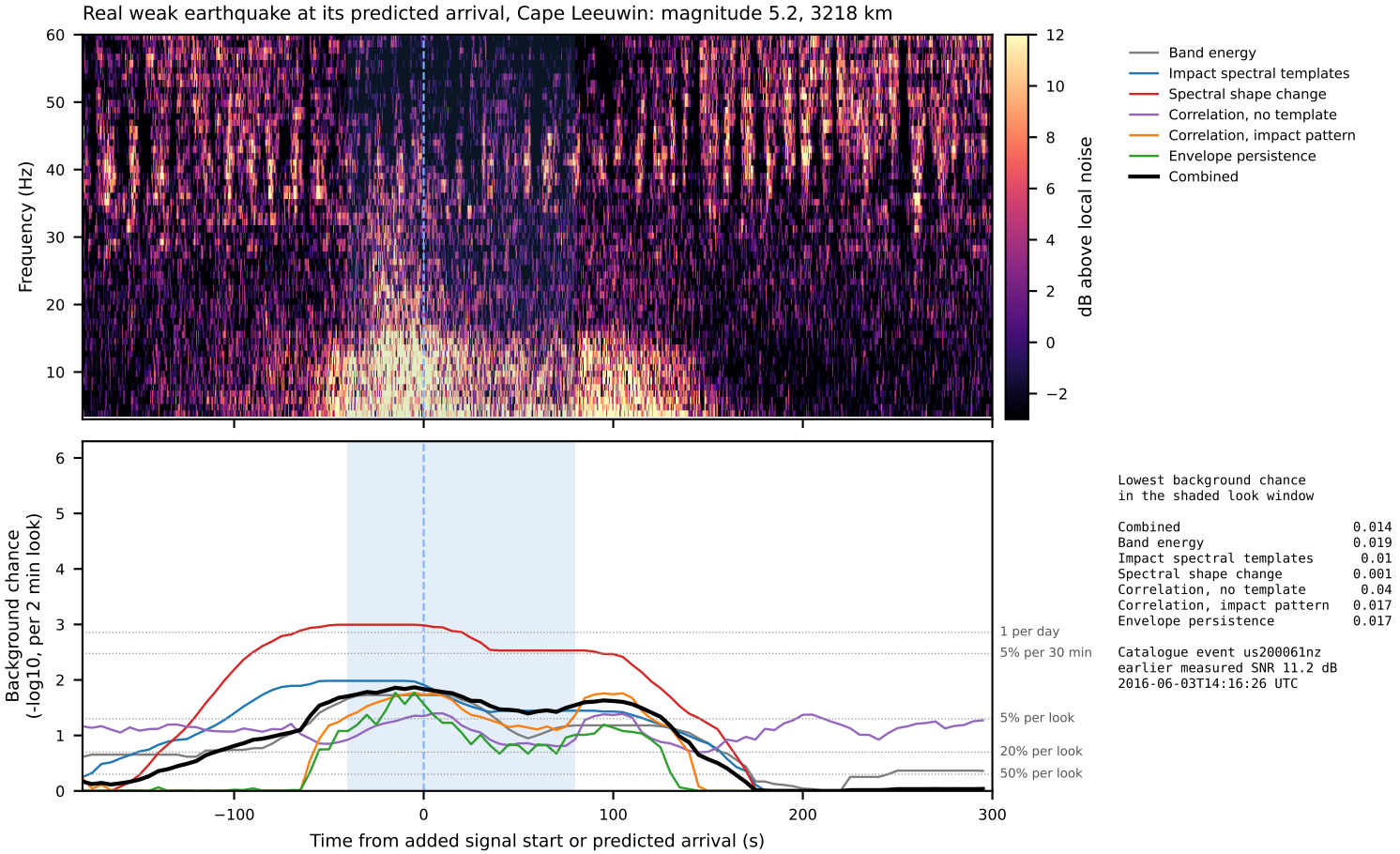
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance in the shaded look window

Combined	0.055
Band energy	0.13
Impact spectral templates	0.099
Spectral shape change	0.026
Correlation, no template	0.093
Correlation, impact pattern	0.057
Envelope persistence	0.086

Catalogue event us20006eqj
earlier measured SNR 6.5 dB
2016-07-14T18:59:20 UTC

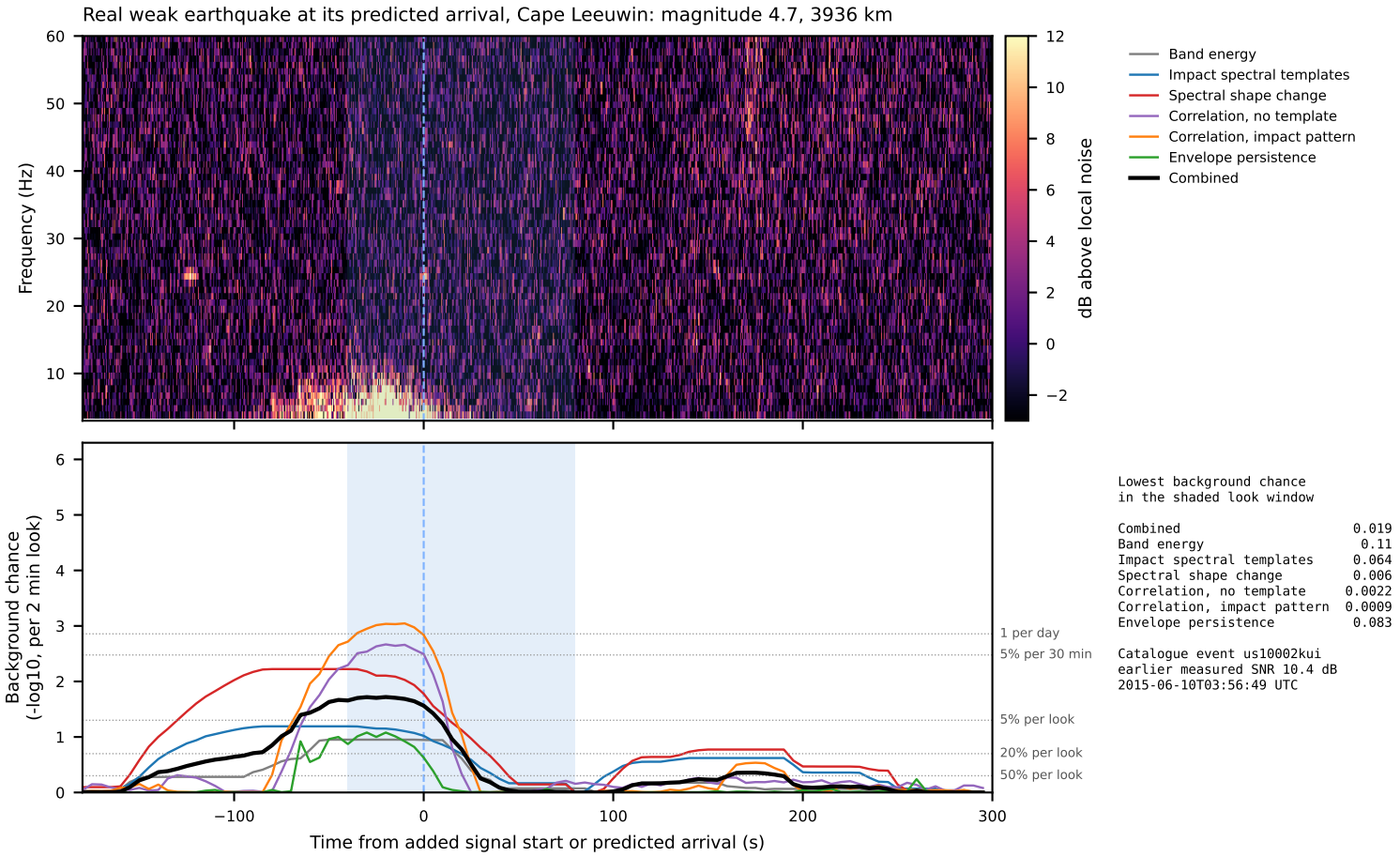


Lowest background chance in the shaded look window

Combined	0.014
Band energy	0.019
Impact spectral templates	0.01
Spectral shape change	0.001
Correlation, no template	0.04
Correlation, impact pattern	0.017
Envelope persistence	0.017

Catalogue event us200061nz
earlier measured SNR 11.2 dB
2016-06-03T14:16:26 UTC

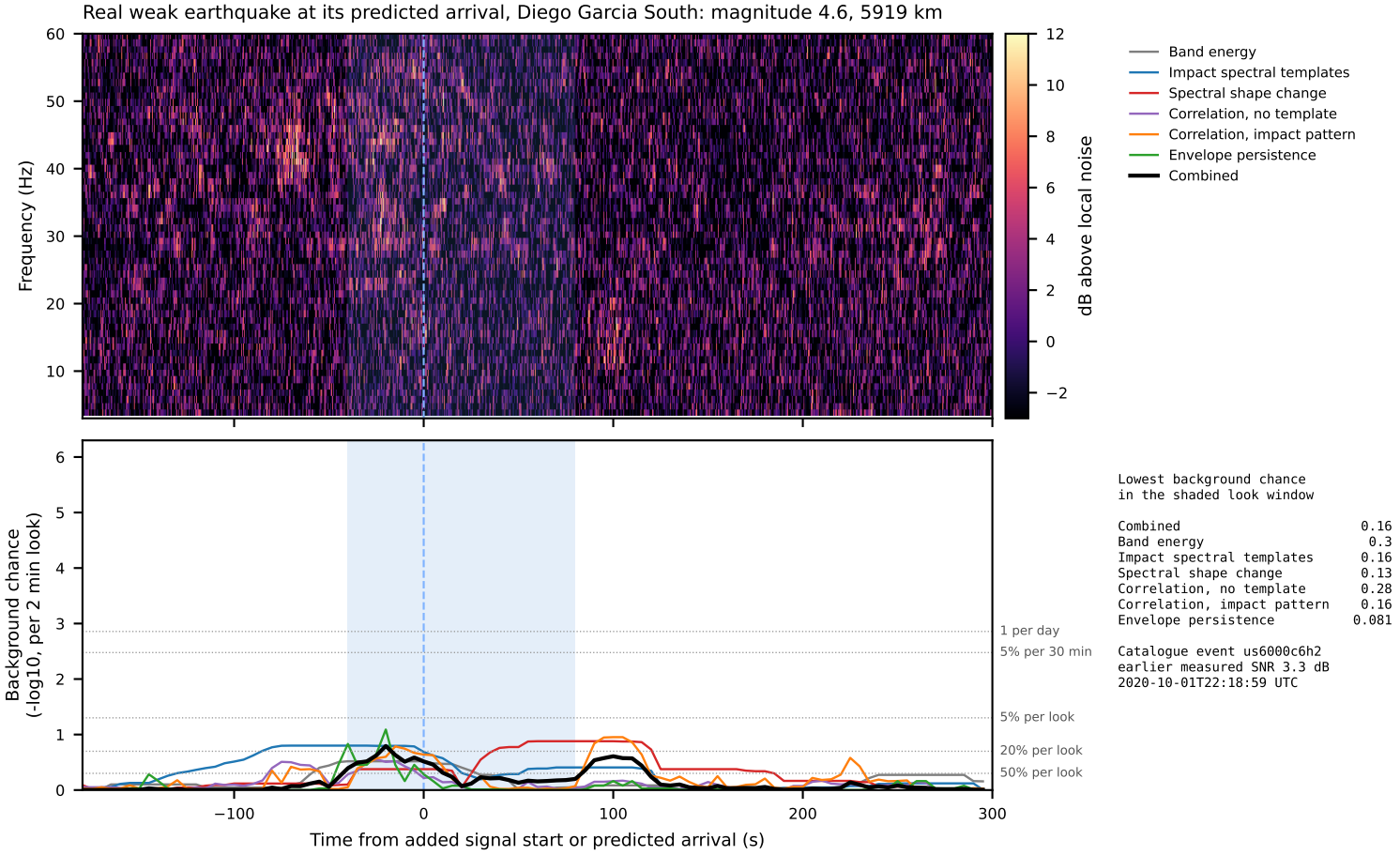
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.019
Band energy	0.11
Impact spectral templates	0.064
Spectral shape change	0.006
Correlation, no template	0.0022
Correlation, impact pattern	0.0009
Envelope persistence	0.083

Catalogue event us10002kui
earlier measured SNR 10.4 dB
2015-06-10T03:56:49 UTC

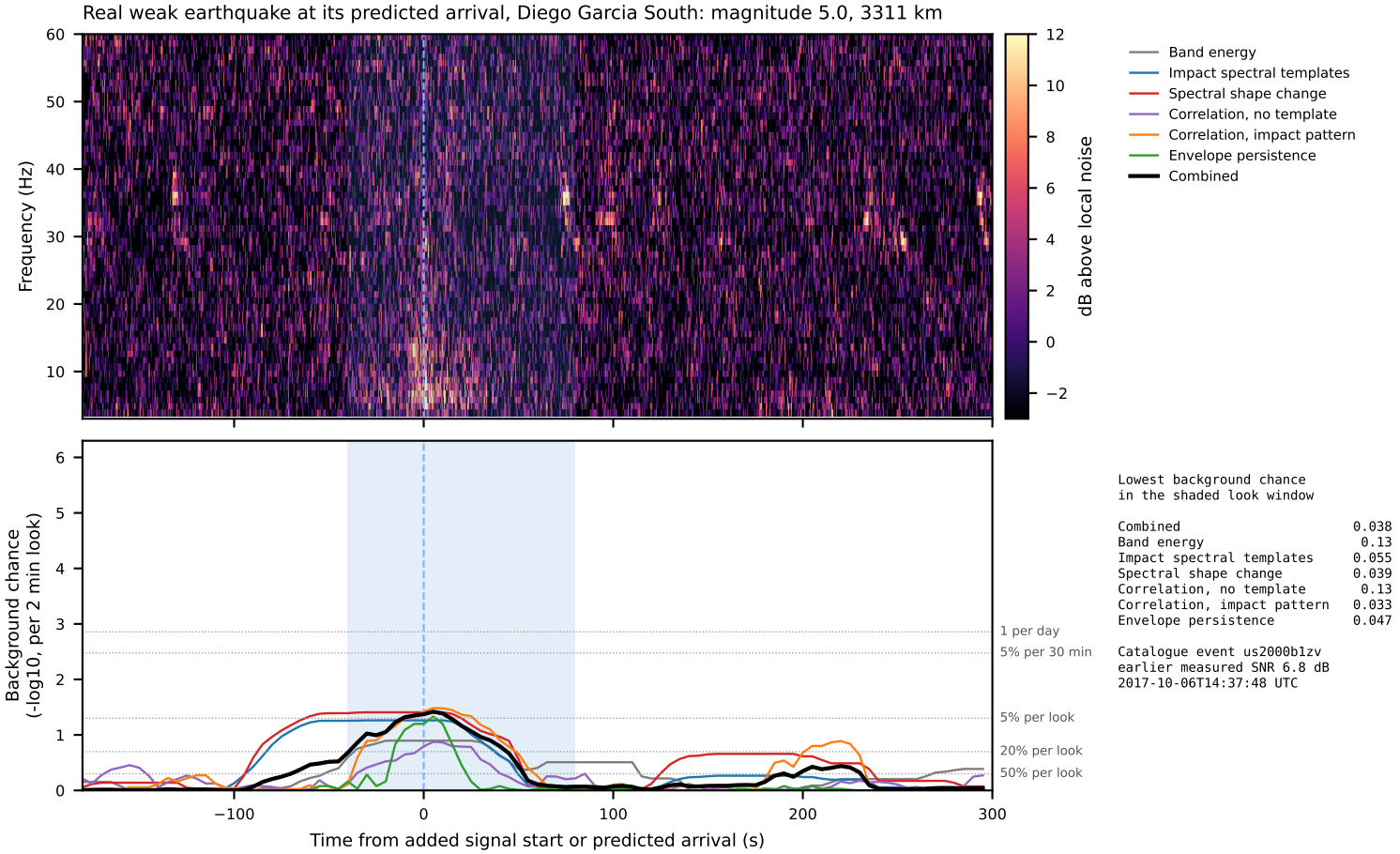


Lowest background chance
in the shaded look window

Combined	0.16
Band energy	0.3
Impact spectral templates	0.16
Spectral shape change	0.13
Correlation, no template	0.28
Correlation, impact pattern	0.16
Envelope persistence	0.081

Catalogue event us6000c6h2
earlier measured SNR 3.3 dB
2020-10-01T22:18:59 UTC

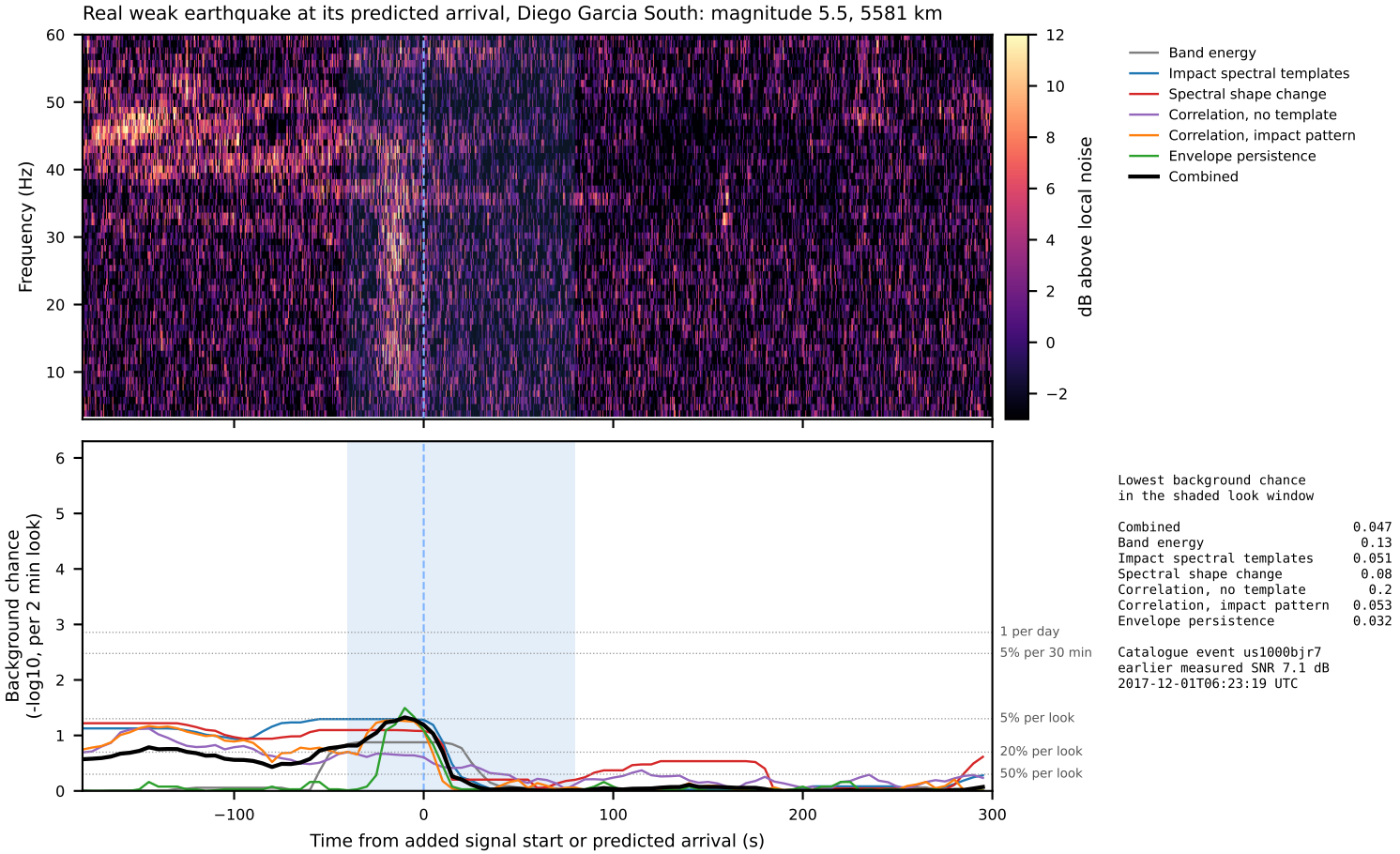
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



Lowest background chance
in the shaded look window

Combined	0.038
Band energy	0.13
Impact spectral templates	0.055
Spectral shape change	0.039
Correlation, no template	0.13
Correlation, impact pattern	0.033
Envelope persistence	0.047

Catalogue event us2000b1zv
earlier measured SNR 6.8 dB
2017-10-06T14:37:48 UTC

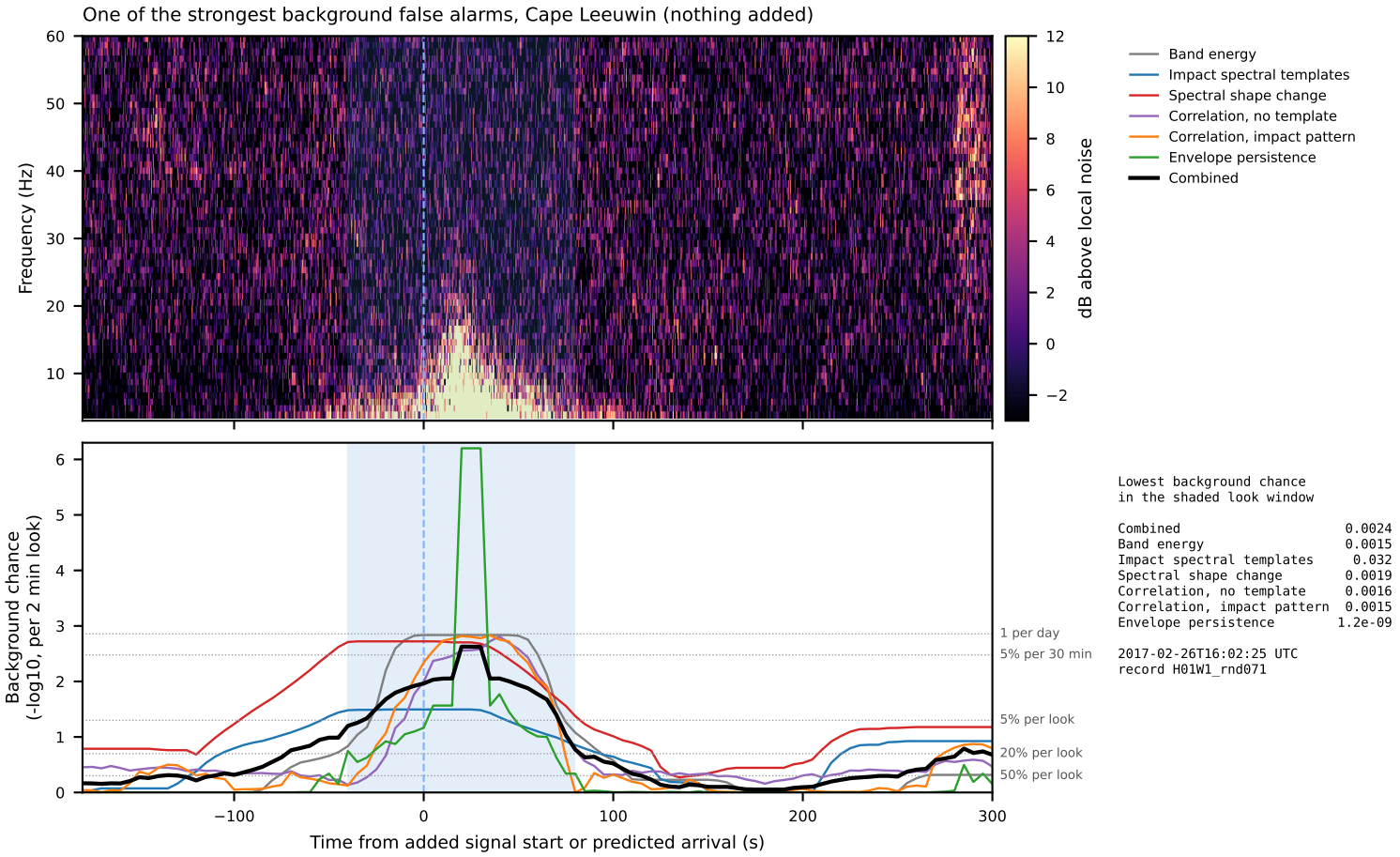


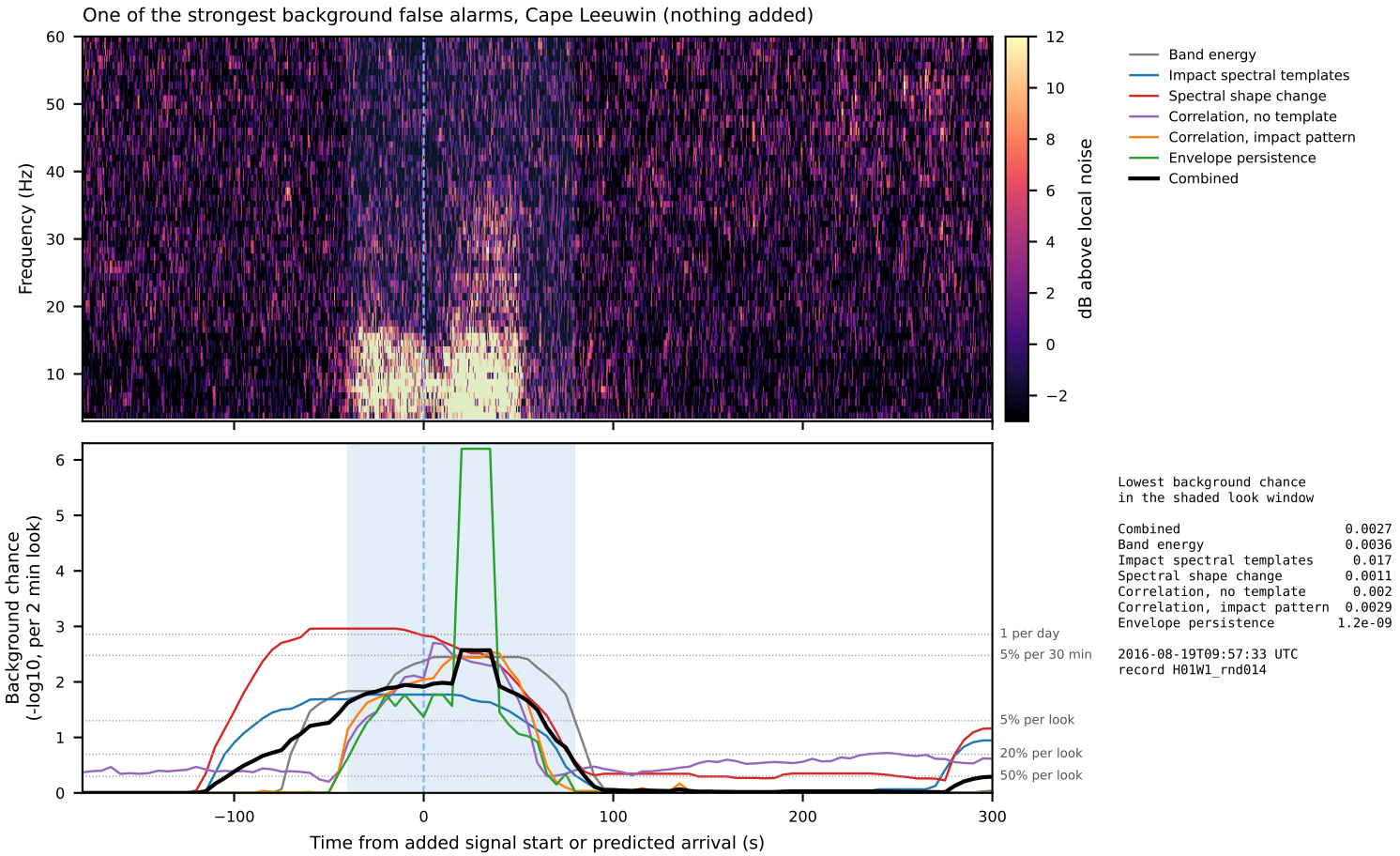
Lowest background chance
in the shaded look window

Combined	0.047
Band energy	0.13
Impact spectral templates	0.051
Spectral shape change	0.08
Correlation, no template	0.2
Correlation, impact pattern	0.053
Envelope persistence	0.032

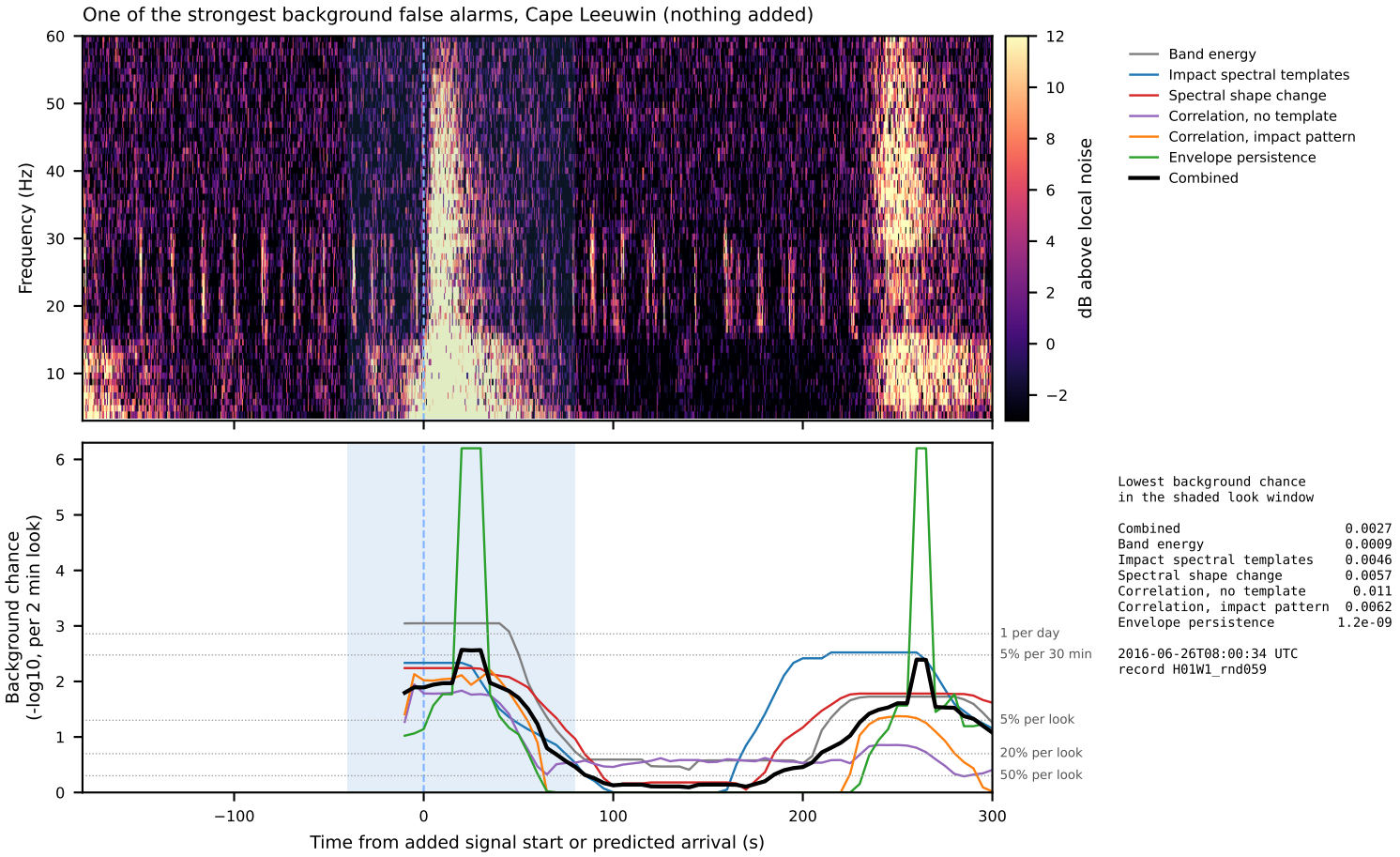
Catalogue event us1000bjr7
earlier measured SNR 7.1 dB
2017-12-01T06:23:19 UTC

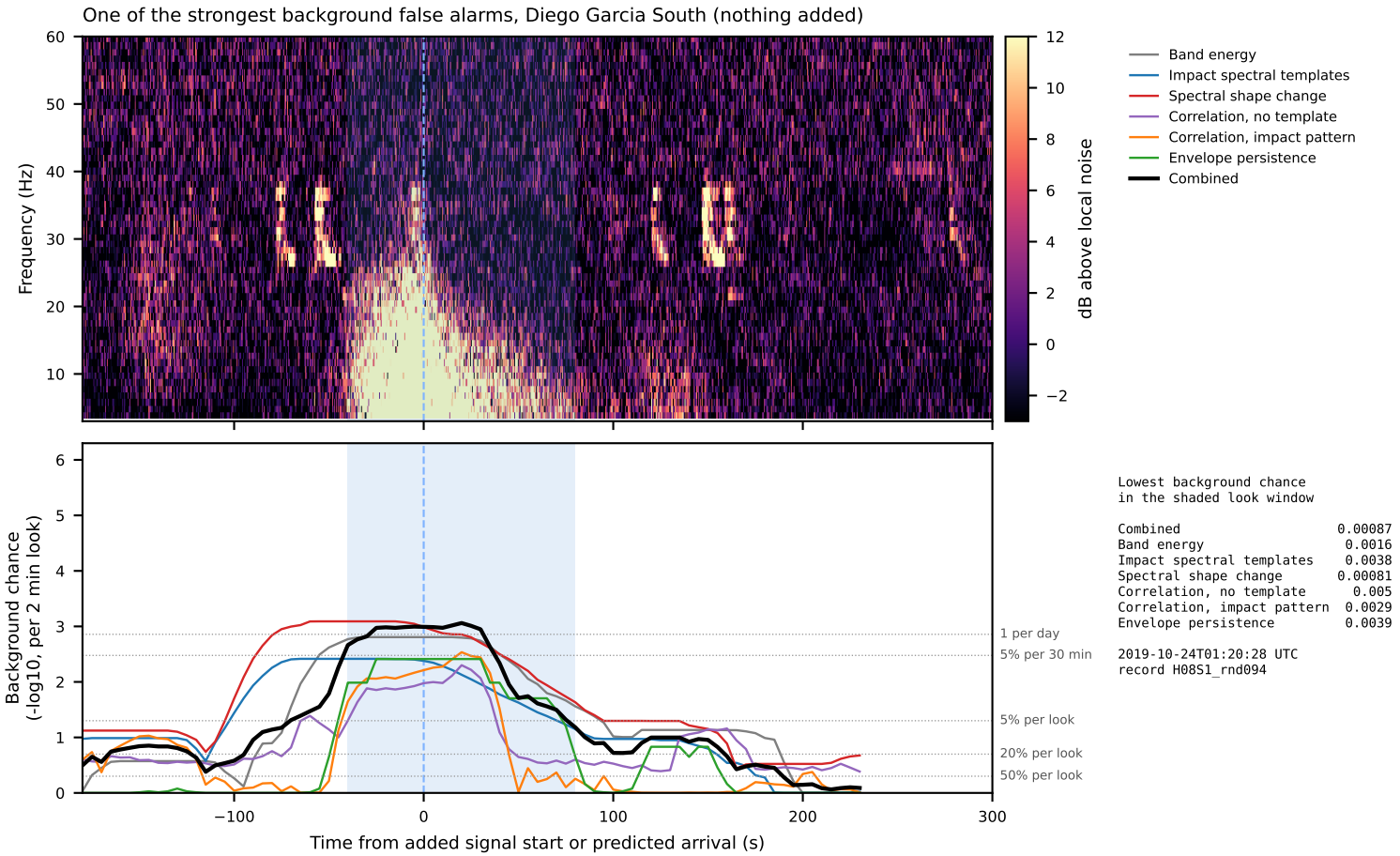
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.



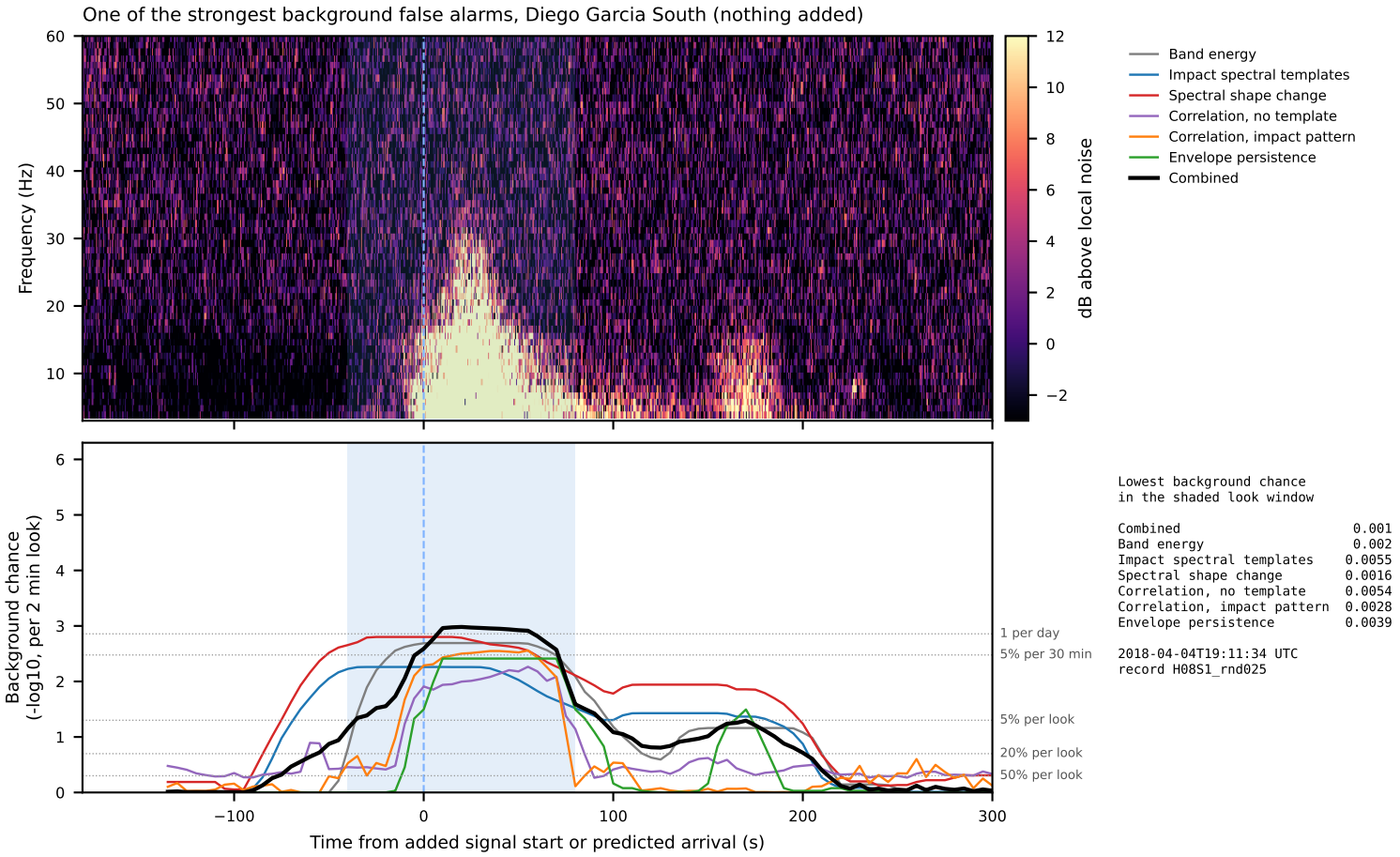


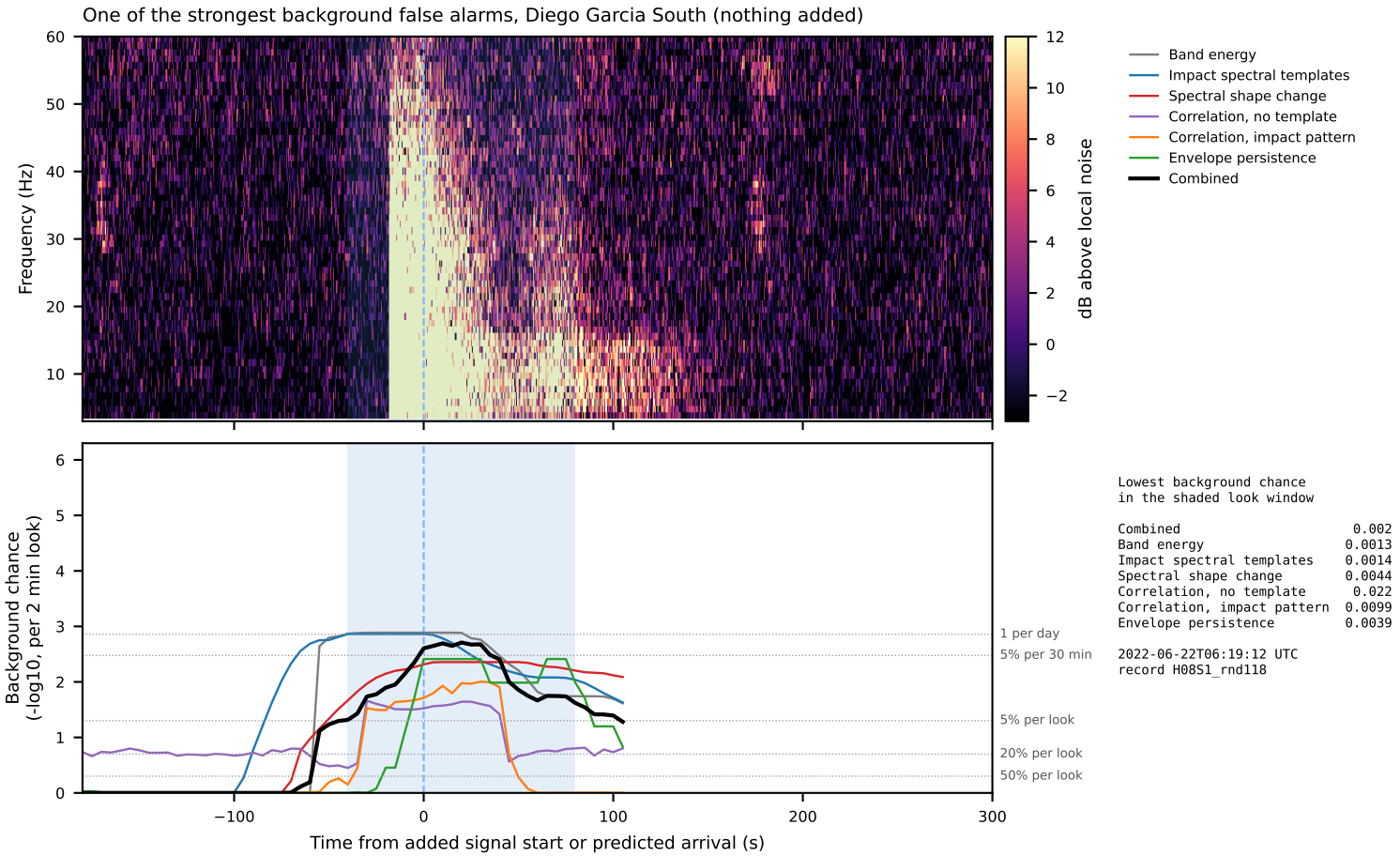
Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.





Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.





Open IMS hydrophone data (EarthScope), element 1, 250 samples/s. Spectrogram: 1 s Hann frames, whitened by a 10 min running median noise spectrum. Lower panel: for each detector, the chance that real background alone gives this score or higher in a 2 min look (144 / 178 random 40 min records, 2015-2025, generalised Pareto tail beyond the 90th percentile). Combined = Fisher sum of band energy, spectral shape change, correlation with the impact pattern and envelope persistence, calibrated on the same background. Shaded: the 2 min look window used for scoring. SNR = largest 30 s signal energy in 5-40 Hz over the local noise energy in 30 s. Synthetic signals: 1,500-4,500 km deep channel, 2-4 Hz low cut-off, Thorp absorption, 1-95 Hz instrument response. Spectral-template scores are tied to the template start, so they rise up to about 90 s before a signal. Provisional methods test; no 8 March 2014 data used.